

The sharp factor-of-IID threshold for the free Ising model on regular trees

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Abstract

For every integer $d \geq 3$ and inverse temperature $\beta \geq 0$, the free zero-field ferromagnetic Ising measure on the infinite d -regular tree is a factor of IID exactly when $\tanh \beta \leq (d-1)^{-1/2}$. We prove the positive implication, including equality, resolving the conjecture of Nam, Sly and Zhang; the strict converse is known. The spin factor can be chosen to commute with every tree automorphism on every label input.

1 Introduction

Can the free Ising state on a regular tree be generated from independent vertex labels without choosing a root? The broadcast description of the state begins at a vertex and propagates a random spin along the edges. Its law does not depend on that vertex. A factor construction must also remove the distinguished vertex from the rule that generates the spins.

Let $T_d = (\mathcal{V}, \mathcal{E})$ be the infinite d -regular tree, $d \geq 3$, and put $b = d-1$ and $\theta = \tanh \beta$ for $\beta \geq 0$. The free zero-field Ising law $\mu_{d,\beta}$ is obtained by choosing a fair spin at one vertex and, independently along edges directed away from it, letting a child agree with its parent with probability $(1+\theta)/2$. Its marginal on any finite connected subtree D is

$$\mu_{d,\beta}(\sigma|_D = \eta) = \frac{1}{2} \prod_{\{u,v\} \in \mathcal{E}(D)} \frac{1 + \theta \eta_u \eta_v}{2}.$$

This expression also verifies that the law is invariant under all tree automorphisms and independent of the chosen broadcast root.

We call a spin law a *factor of IID* if it is the law of $\Phi(U)$ for independent uniform labels $(U_v)_{v \in \mathcal{V}}$ and a measurable map $\Phi : [0, 1]^\mathcal{V} \rightarrow \{-1, 1\}^\mathcal{V}$ satisfying

$$\Phi(gU) = g\Phi(U) \quad \text{almost surely for each fixed } g \in \text{Aut}(T_d), \quad (gx)_v = x_{g^{-1}v}. \quad (1.1)$$

The exceptional null set may depend on g . For spin laws this convention is equivalent to requiring a Borel factor equivariant on every input: Lemma 8.5 constructs such a version. The input contains no distinguished root or end. Measurability does not require a spin to be determined by finitely many labels; no finitary coding or coding-radius estimate is asserted.

Theorem 1.1. *For every integer $d \geq 3$ and every $\beta \geq 0$, the free zero-field Ising law $\mu_{d,\beta}$ is a factor of IID in the sense of (1.1) if and only if*

$$\tanh \beta \leq (d-1)^{-1/2}.$$

In this range the factor can be chosen to commute with every automorphism on every label input.

The construction proves the positive implication, including equality. The strict converse is known: the obstruction beyond the reconstruction threshold is due to Sly, as recorded and proved in Lyons [12, Section 3 and Corollary 3.2]. We use the quantitative correlation bound of Backhausz, Szegedy and Virág [1, Theorem 3.1 in arXiv:1305.6784v1] to give that implication below. The map is obtained for each fixed d and β in the permitted range; no parameter-uniform or computational bound is asserted.

For the free state on T_d , reconstruction from distant levels is possible exactly when $b\theta^2 > 1$; equivalently, the free Gibbs state is extremal exactly when $b\theta^2 \leq 1$. For the regular-tree result, including equality, see the Bethe-lattice work of Bleher, Ruiz, and Zagrebnov [3] and Ioffe [9]. Ioffe [10] and Evans, Kenyon, Peres, and Schulman [6] developed branching-number criteria on general trees away from equality. Pemantle and Peres [17, Theorem 2.1] later gave an electrical-capacity criterion for free-state extremality for couplings bounded above and away from zero, which also covers critical cases.

Reconstruction asks whether distant spins retain information about a fixed spin. The factor question asks for a rule that generates the entire law from independent labels. Nonreconstruction supplies no such rule, and a weak limit of factor laws need not itself be a factor [12, Corollary 3.3]. Thus equality requires a construction that retains measurable dependence on its independent input.

In the uniqueness range $\theta \leq b^{-1}$, including equality, a finite-cluster construction gives the factor; see Lyons [12, Section 3]. Nam, Sly and Zhang [14, Theorem 1 and Section 2.2 in arXiv:2012.09484v2] used posterior-drift stochastic differential equations driven by IID Brownian motions to construct the free Ising law as a factor for $d \geq d_0$ and $\theta \leq c/\sqrt{d-1}$, for unspecified absolute constants $c, d_0 > 0$. Their construction takes root-independent limits of finite-volume solutions. Theorem 1.1 proves their conjecture for every $d \geq 3$, including equality and their pointwise convention for equivariance.

The known converse. For a centered variance-one real vertex factor, the BSV bound is

$$|\mathbb{E}[Z_o Z_v]| \leq \left(1 + \frac{d-2}{d}n\right) b^{-n/2}, \quad \text{dist}(o, v) = n.$$

It applies also under (1.1). To check this, condition the observable at o on the labels in its radius- R ball and average a Borel version over the finite automorphism group of that rooted ball. Every such permutation extends to a tree automorphism fixing o , so uniqueness of conditional expectation leaves this average unchanged almost surely. It now defines an exactly invariant local rule. Apply the same rule at every vertex. These local factors converge in L^2 at each vertex to the original factor, because the balls exhaust the label coordinates. For all sufficiently large R their variances are positive. Center and normalize them, apply BSV, and pass to the limit in variances and covariances on this same IID probability space. Since the Ising spins have mean zero, variance one and correlation θ^n , the bound would require $(\theta\sqrt{b})^n \leq 1 + (d-2)n/d$. This is impossible when $\theta > b^{-1/2}$.

1.1 Why independent innovations determine the spins

We follow the observation framework of Nam–Sly–Zhang. On an auxiliary probability space, sample the desired spin configuration σ and independent standard Brownian motions B_v , and observe

$$X_v(t) = t\sigma_v + B_v(t), \quad m_v(t) = \mathbb{E}[\sigma_v \mid X_u(s) : u \in \mathcal{V}, 0 \leq s \leq t].$$

The finite-dimensional posterior evolution also belongs to stochastic localization. Eldan [5] developed the underlying localization scheme, and El Alaoui and Montanari [4, Section 3, Theorem 2] give the

fixed-covariance Gaussian-observation formulation used here. Subtracting the conditional drift gives

$$W_v(t) = X_v(t) - \int_0^t m_v(s) ds.$$

These innovations are independent Brownian motions in the observation filtration. This property alone does not make X a function of W : the original construction still used the spins. Classical innovations theory separates the Brownian property from generation of the observation filtration [7, Lemma 2.2 and Remark 2.2]. Proving that the innovations determine X is the central step here.

A companion construction for the free uniform spanning forest uses a uniform Lipschitz response bound in the coordinate supremum norm to invert the innovations causally for every continuous input [15, Section 4]. Here recovery instead follows from equality of conditional copies under the observation law.

Take two observations X, X' independently conditional on the entire innovation array W . They have the same marginal law and share the same innovations. Conditioning on the full paths could in principle expose their future increments through an observation past. We prove that it does not: the conditional law of either observation past given all of W depends only on the past of W . Hence W remains Brownian in the joined filtration, and the equations for both observations use the same driving Brownian motions. This is the compatibility principle for conditional copies [11, Lemmas 2.11–2.12 in arXiv:1305.6747v2]; we prove the required statement and the transport of the stochastic integrals directly.

We then show that these particular copies agree. For each coordinate, conditional independence gives

$$\mathbb{E}[(X_v(t) - X'_v(t))^2 \mid W] = 2 \operatorname{Var}(X_v(t) \mid W).$$

Equality therefore forces the conditional law of X given W to be a point mass. Finally

$$\frac{X_v(n)}{n} \longrightarrow \sigma_v \quad \text{almost surely}$$

recovers the spins from W . A common measurable sampler converts one uniform label at each vertex into its Brownian path. This gives a spin factor satisfying (1.1); finite-ball averaging then selects an everywhere equivariant version with the same law.

1.2 What changes at the critical point

The tree gives a concrete equation for comparing the copies. Set $h_v = \operatorname{atanh} m_v$. If $T_{j \rightarrow i}$ is the component containing j after deleting ij , define its field by

$$h_{j \rightarrow i}(t) = \operatorname{atanh} \mathbb{E}[\sigma_j \mid X_u(s) : u \in T_{j \rightarrow i}, 0 \leq s \leq t],$$

using the unconditional prior on that component. For a path $v = v_0, \dots, v_r = u$, the field-response coefficient is

$$c_{vu} = \theta^r \prod_{i=1}^r a(h_{v_i \rightarrow v_{i-1}}), \quad a(z) = \frac{1 - \tanh^2 z}{1 - \theta^2 \tanh^2 z}, \quad c_{vv} = 1.$$

The innovation equation is

$$dh_v = \sum_u c_{vu} dW_u + m_v A_v dt, \quad A_v = \sum_u c_{vu}^2.$$

Its infinite sum requires proof. At criticality we obtain $\mathbb{E}A_v(t) \leq C/\min(\sqrt{t}, 1)$ for $t > 0$, so the row is integrable down to time zero. An exact covariance-energy identity and a testing argument establish the equation without a residual martingale.

For the conditional copies, let $V(t) = \mathbb{E}[(h_o(t) - h'_o(t))^2]$; invariance makes the choice of o irrelevant. It starts at zero. Below the threshold, the squared factor θ^{2r} and the shell size db^{r-1} give a summable geometric series, yielding $V' \leq CV$ and thus equality of the copies. At $b\theta^2 = 1$ those two factors cancel exactly.

Three estimates replace the lost geometric decay. First, an intact side branch attenuates a path product on the length scale $t^{-1/2}$ at small times. Second, spatial cancellation bounds the variance of a sum of differences between the two copies' $\log a(h_{j \rightarrow i})$ factors by the path length times the variance at one edge. The spectral bound is proved for differences $f(X) - f(X')$, with the same square-integrable equivariant vertex observable f applied to each copy, and their closed linear span. No spectral bound for every function of the pair is assumed. The operator argument is related to the correlation methods of Backhausz–Szegedy–Virág [1] and Backhausz–Gerencsér–Harangi–Vizer [2]. Third, concentration of path averages makes the attenuation usable with discrepancy weights that depend on the same observations. Both path orientations are needed in this averaging argument.

These estimates leave one small-time difficulty: the leading noise term is of order $S(t)/t$, where

$$S(t) = \mathbb{E}[(\log a(h_{j \rightarrow i}(t)) - \log a(h'_{j \rightarrow i}(t)))^2].$$

The elementary bound $S \leq CV$ would leave a nonintegrable coefficient. We prove instead a small-field tail estimate and use the vanishing derivative of $\log a$ at zero to improve that comparison. The complete noise and drift calculation then gives an integrable Osgood-type inequality for V , forcing $V = 0$. The moment estimate is proved beside this final comparison, where its purpose becomes explicit.

Notation and organization. The case $\theta = 0$ is already IID, so the proof below assumes $0 < \theta \leq b^{-1/2}$. Constants $c, C > 0$ may depend on the fixed d, θ and may change between occurrences; a subscript T permits additional dependence on a fixed finite time horizon. Every path is simple unless specified otherwise, and all logarithms are natural. Roots used to describe subtrees or compute norms are auxiliary choices in the proof, not inputs to the factor.

Sections 2–4 develop the posterior identities, critical information bounds, and infinite innovation equation. Section 5 constructs the conditional copies and proves their equality below the threshold. Sections 6 and 7 establish spatial cancellation and concentration of path averages. Section 8 proves the small-field tail estimate, completes the critical comparison, and recovers the factor from the innovations.

2 Posterior fields along the tree

The observation construction reduces the factor problem to a question about conditional spin laws. We first describe those laws on a finite subtree, where the Gaussian likelihood is explicit. The tree structure then expresses each covariance as a product along the connecting path. Passing these formulas to the whole tree will identify the response coefficients whose squared rows must be integrable in time.

For the affirmative implication, write $b = d - 1$ and assume $0 < \theta = \tanh \beta \leq b^{-1/2}$; the case $\theta = 0$ is already IID. On an auxiliary probability space sample the free Ising configuration σ , and independently sample standard Brownian motions $(B_v)_{v \in \mathcal{V}}$, independent across vertices. Set

$$X_v(t) = t\sigma_v + B_v(t), \quad \mathcal{F}_t^0 = \sigma\{X_v(s) : v \in \mathcal{V}, 0 \leq s \leq t\}. \quad (2.1)$$

The B_v 's are the noise used to plant these observations. The innovations W_v , obtained later by subtracting the conditional drift from X_v , are different processes. In this section $\mathcal{F}_t^0$ is the raw

observation filtration; we reserve $\mathcal{F}_t$ for its usual augmentation. For any $D \subseteq \mathcal{V}$, write

$$\mathcal{F}_t^D = \sigma\{X_u(s) : u \in D, 0 \leq s \leq t\}.$$

The Gaussian observation framework follows Nam–Sly–Zhang [14]. In finite volume it is the observation form of stochastic localization described by El Alaoui–Montanari [4, Section 3]. We derive the finite posterior directly, since its derivative and its tree factorization will both be needed.

2.1 The finite posterior and its branch messages

Let D be a finite connected subtree. Its broadcast marginal is

$$\mathbb{P}(\sigma_D = \eta) = 2^{-|D|} \prod_{\{i,j\} \in \mathcal{E}(D)} (1 + \theta \eta_i \eta_j) \propto \exp \left(\beta \sum_{\{i,j\} \in \mathcal{E}(D)} \eta_i \eta_j \right), \quad \eta \in \{-1, 1\}^D.$$

Indeed, choose any vertex of D as the broadcast root and multiply its probability $1/2$ by the transition probabilities on the $|D| - 1$ edges; also $1 + \theta s = e^{\beta s} / \cosh \beta$ for $s \in \{-1, 1\}$. Conditional on $\sigma_D = \eta$, the likelihood of the observation paths through time t , relative to product Wiener measure, is

$$\exp \left(\sum_{u \in D} \eta_u X_u(t) - \frac{t|D|}{2} \right).$$

The last term is independent of η . Bayes' formula gives, for every finite terminal vector $x \in \mathbb{R}^D$,

$$\pi_x^D(\eta) = \frac{1}{Z_D(x)} \exp \left(\beta \sum_{\{i,j\} \in \mathcal{E}(D)} \eta_i \eta_j + \sum_{u \in D} x_u \eta_u \right), \quad x = X_D(t). \quad (2.2)$$

Here $Z_D(x)$ is the positive finite partition function. This calculation shows that the terminal vector is sufficient for the spin posterior; it does not identify the terminal-vector sigma-field with the entire observation history.

Let

$$M_v^D(x) = \sum_{\eta} \eta_v \pi_x^D(\eta), \quad m_v^D(t) = M_v^D(X_D(t)), \quad h_v^D(t) = \operatorname{atanh} m_v^D(t).$$

By Bayes' formula, $m_v^D(t)$ is a version of $\mathbb{E}[\sigma_v \mid \mathcal{F}_t^D]$. All posterior masses are positive, so $|M_v^D(x)| < 1$ and the field is finite. The functions M_v^D are smooth. If $u, v \in D$, then differentiating the finite sum in (2.2) gives

$$\frac{\partial M_v^D}{\partial x_u}(x) = \operatorname{Cov}_{\pi_x^D}(\eta_v, \eta_u). \quad (2.3)$$

At $x = X_D(t)$ we denote this conditional covariance by $C_{vu}^D(t)$.

For an oriented edge $v \rightarrow w$, let $T_{v \rightarrow w}$ be the component containing v after deletion of $\{v, w\}$. If both vertices belong to D , define

$$h_{v \rightarrow w}^D(t) = h_v^{D \cap T_{v \rightarrow w}}(t).$$

Thus a component field uses only its own component's observations and the unconditional free prior on that component. It is not the field at v after observing the opposite side or specifying the spin at w . The same distinction will be essential on the infinite tree.

Lemma 2.1 (Posterior identities). *For a fixed deterministic time $t \geq 0$, set*

$$m_v(t) = \mathbb{E}[\sigma_v \mid \mathcal{F}_t^0], \quad m_{v \rightarrow w}(t) = \mathbb{E}[\sigma_v \mid \mathcal{F}_t^{T_{v \rightarrow w}}].$$

These means lie in $(-1, 1)$ almost surely. Write $h_v = \operatorname{atanh} m_v$ and $h_{v \rightarrow w} = \operatorname{atanh} m_{v \rightarrow w}$, and define

$$\psi(z) = \operatorname{atanh}(\theta \tanh z), \quad a(z) = \frac{1 - \tanh^2 z}{1 - \theta^2 \tanh^2 z}.$$

At each fixed time, almost surely,

$$\begin{aligned} h_{v \rightarrow w}(t) &= X_v(t) + \sum_{\substack{u \sim v \\ u \neq w}} \psi(h_{u \rightarrow v}(t)), \\ h_v(t) &= X_v(t) + \sum_{u \sim v} \psi(h_{u \rightarrow v}(t)) = h_{v \rightarrow w}(t) + \psi(h_{w \rightarrow v}(t)). \end{aligned} \tag{2.4}$$

The fields have causal, progressively measurable Borel versions as functions of the observation paths. The component version uses only paths in $T_{v \rightarrow w}$. These versions require no distinguished root and satisfy, at every input and time,

$$|h_v(t) - X_v(t)| \leq d\beta, \quad |h_{v \rightarrow w}(t) - X_v(t)| \leq b\beta. \tag{2.5}$$

In the process formulas below, use these chosen field versions and set $m_v(t) = \tanh h_v(t)$ and $m_{v \rightarrow w}(t) = \tanh h_{v \rightarrow w}(t)$. For the simple path $v = v_0, v_1, \dots, v_r = u$, define

$$\begin{aligned} c_{vu}(t) &= \theta^r \prod_{i=1}^r a(h_{v_i \rightarrow v_{i-1}}(t)), \quad c_{vv}(t) = 1, \\ C_{vu}(t) &= (1 - m_v(t)^2) c_{vu}(t). \end{aligned} \tag{2.6}$$

The last expression is a progressively measurable version of $\operatorname{Cov}(\sigma_v, \sigma_u \mid \mathcal{F}_t^0)$ at each fixed time. The cavity and covariance formulas also hold in every finite connected subtree, with branches restricted to that subtree. For all real z ,

$$0 < a(z) \leq 1, \quad |\psi(z)| \leq \beta, \quad \psi'(z) = \theta a(z) \in (0, \theta],$$

and $0 \leq C_{vu}(t) \leq \theta^{\operatorname{dist}(v, u)}$ at each fixed time.

Proof. We first establish the identities in a finite connected subtree D . A branch whose root has field z assigns that root probabilities proportional to e^z and e^{-z} . Summing its root spin against the edge interaction gives a factor proportional to $\cosh(z + \beta s)$ when its parent spin is s . Its contribution to the parent's half log-odds is therefore

$$\frac{1}{2} \log \frac{\cosh(z + \beta)}{\cosh(z - \beta)} = \operatorname{atanh}(\theta \tanh z) = \psi(z).$$

The branch likelihoods multiply because the components are disjoint conditional on the parent spin. Adding their half log-odds and the own observation $X_v(t)$ proves the finite cavity formulas. Since $|\psi| \leq \beta$, the site field differs from $X_v(t)$ by at most $d\beta$, and a component field by at most $b\beta$.

We next follow a path $v_0, \dots, v_r$ in D . With the observations fixed, sum the posterior in (2.2) over all spins off the path. Given the preceding path spins, the conditional mean of the spin at v_i uses only $\sigma_{v_{i-1}}$ and the field on the side excluding that predecessor:

$$\mathbb{E}[\sigma_{v_i} \mid \sigma_{v_0}, \dots, \sigma_{v_{i-1}}, \mathcal{F}_t^D] = \tanh(z + \beta \sigma_{v_{i-1}}) = p_i + q_i \sigma_{v_{i-1}}, \quad z = h_{v_i \rightarrow v_{i-1}}^D(t).$$

Here p_i is an intercept depending on the observations, and

$$q_i = \frac{\tanh(z + \beta) - \tanh(z - \beta)}{2} = \theta \frac{1 - \tanh^2 z}{1 - \theta^2 \tanh^2 z} = \theta a(z).$$

Iterating these affine conditional means makes the slope of the conditional mean at v_r , as a function of σ_{v_0} , equal to $\prod_{i=1}^r q_i$. Taking covariance with the first spin gives

$$C_{v_0 v_r}^D(t) = (1 - (m_{v_0}^D(t))^2) \prod_{i=1}^r q_i.$$

This proves the finite path formula, including the empty product at $r = 0$. The coefficient c_{vu}^D is directed even though the covariance is symmetric: its normalization uses the variance at v . In fact, (2.3) gives

$$\frac{\partial}{\partial x_u} \operatorname{atanh} M_v^D(x) = \frac{\operatorname{Cov}_{\pi_x^D}(\eta_v, \eta_u)}{1 - M_v^D(x)^2},$$

which is the path product just obtained. Thus c_{vu}^D is the finite field's response to the observation at u .

Now fix t and exhaust the tree by increasing finite balls. The conditional expectations of a fixed spin, or a product of two fixed spins, converge almost surely and in L^2 to conditioning on $\mathcal{F}_t^0$. This is increasing-sigma-field convergence for bounded random variables. Intersecting the balls with $T_{v \rightarrow w}$ gives the corresponding convergence for the component posterior. The finite field bounds place each mean in the random compact interval

$$[\tanh(X_v(t) - k\beta), \tanh(X_v(t) + k\beta)] \subset (-1, 1), \quad k = d \text{ or } b.$$

Thus the limiting means remain strictly inside $(-1, 1)$, and their inverse hyperbolic tangents converge to finite fields. For a fixed path, take balls large enough to contain it. Their intersections with the finitely many directed components on the path exhaust those components. The finite cavity sums and path product therefore pass to the limit; convergence of the two-spin moments identifies the limit as the conditional covariance. Any other centered exhaustion gives the same conditional expectations, since its histories generate the same sigma-fields.

It remains to choose process versions without presupposing a continuous infinite-dimensional filter. On the canonical space of continuous observation paths, evaluate the finite field formulas on $B_R(v)$ for a site and on $B_R(v) \cap T_{v \rightarrow w}$ for a component, with $R = 0, 1, \dots$. Define each raw field to be the pointwise $\limsup_{R \rightarrow \infty}$ of these finite fields. Each finite expression is a continuous adapted terminal-array function. Its upper limit is Borel and progressively measurable, for the whole observation filtration or for the specified component filtration respectively. The finite bounds make the upper limits finite and preserve (2.5) at every input and time. At each fixed time they agree almost surely with the conditional fields proved above. Their hyperbolic tangents and the finite products in (2.6) give the claimed mean and covariance versions.

These formulas use only the terminal array at the evaluated time, and in particular only the histories up to that time. Centering every ball at its indexed vertex makes the formulas commute with graph relabeling. Joint measurability and Fubini's theorem turn the fixed-time algebraic identities into identities for almost every time, almost surely. Countability of the vertices, oriented edges, finite paths, and integer time horizons permits all these time-probability statements to hold simultaneously. Continuity and identities simultaneous at every time are not needed here; they will be obtained after the filtering argument in Section 4.

Finally, direct differentiation gives $\psi'(z) = \theta(1 - \tanh^2 z)/(1 - \theta^2 \tanh^2 z)$. Since $0 < \theta < 1$, its bounds and $0 < a \leq 1$ follow. The covariance bound follows from the finite product and $1 - m_v^2 \leq 1$. $\square$

The normalized response c_{vu} , rather than the covariance alone, is the coefficient that will occur in the field equation. At distance r , its deterministic bound is θ^r . Below the threshold this already gives a summable squared row. At equality, the size of a distance shell cancels θ^{2r} ; the next section obtains the additional decay from the observations in side branches.

3 Critical information controls the response row

At the critical parameter, the squared edge factor in (2.6) is exactly balanced by the number of vertices at that distance. To sum a response row, we must therefore obtain decay from the factors $a(h_{v_i \rightarrow v_{i-1}})$. We will show that the observations in one intact infinite side branch supply a loss of order $\sqrt{t}$ at each path step for small t . The resulting expected squared row is of order at most $t^{-1/2}$, which is integrable at the initial time.

Until the final row estimate, assume $b = d - 1 \geq 2$ and $\theta = b^{-1/2}$. Constants in the critical estimates depend only on the fixed degree d . We work at a fixed deterministic time $t > 0$ with the raw posterior versions from Lemma 2.1; their continuity is not needed. For an oriented edge $v \rightarrow w$, put

$$Y = h_{v \rightarrow w}(t), \quad M(t) = \mathbb{E} \tanh^2 Y, \quad \omega(t) = \min\{\sqrt{t}, 1\}.$$

The law of Y is independent of the oriented edge. It is symmetric: reversing all spins and all observations preserves their joint law and negates the conditional field almost surely at this fixed time. We first determine the scale of $M(t)$. Its lower bound gives path decay, while the whole-site upper bound will control the early-time energy in the filtering limit.

3.1 Combining symmetric binary signals

Combining two branches loses a definite amount of second-moment information compared with adding their separate contributions. This quadratic loss is related to Ioffe's recursive second-moment estimates for Ising trees [10, Proposition 4]. The following elementary formulation will allow us to keep the loss explicit.

Lemma 3.1 (Symmetric binary-signal fusion). *Let ε be a uniform sign and let two random elements of standard Borel spaces be independent conditional on ε . Let U, V be the posterior means of ε from the respective signals. Suppose that their marginal laws are symmetric and $|U|, |V| < 1$ almost surely. If $\mathbb{E}_{\text{prod}}$ denotes expectation under the product of these two marginal laws, then the posterior mean from both signals is $(U + V)/(1 + UV)$, and its second moment is*

$$\mathbb{E}U^2 + \mathbb{E}V^2 - \mathbb{E}_{\text{prod}} \frac{U^2 V^2 (2 - U^2 - V^2)}{1 - U^2 V^2}. \quad (3.1)$$

In particular this second moment is at most $\mathbb{E}U^2 + \mathbb{E}V^2$. If $|U|, |V| \leq \theta$, the subtracted term is at least $2(1 - \theta^2)\mathbb{E}U^2 \mathbb{E}V^2$.

Proof. For a signal with marginal law ν and posterior mean u , the conditional signal law given $\varepsilon = \eta$ has density $1 + \eta u$ with respect to ν , for $\eta \in \{-1, 1\}$. Indeed, condition the indicator of $\{\varepsilon = \eta\}$ on the signal and use its posterior probability $(1 + \eta u)/2$. Conditional independence then gives joint marginal density $1 + UV$ relative to the product of the two signal laws. The signed difference of

the two conditional densities gives the combined posterior mean $(U + V)/(1 + UV)$. Pushing the product signal law forward to the product of the marginal laws of U, V , its second moment is

$$\mathbb{E}_{\text{prod}} \frac{(U + V)^2}{1 + UV}.$$

The integrand is at most 2, so it is legitimate to average it under the symmetry $V \mapsto -V$. The result is

$$\frac{U^2 + V^2 - 2U^2V^2}{1 - U^2V^2} = U^2 + V^2 - \frac{U^2V^2(2 - U^2 - V^2)}{1 - U^2V^2},$$

which proves (3.1). The loss is nonnegative. Under the additional bound by θ , its numerator factor $2 - U^2 - V^2$ is at least $2(1 - \theta^2)$ and its denominator is at most one. The expectation of U^2V^2 under the product law is $\mathbb{E}U^2 \mathbb{E}V^2$, proving the final assertion. $\square$

The subadditivity in this lemma can be iterated over the branches of a vertex. A collection of already combined signals remains independent of every unused signal conditional on the spin. Its posterior mean remains symmetric, and remains strictly inside $(-1, 1)$, since

$$1 - \left(\frac{u + v}{1 + uv} \right)^2 = \frac{(1 - u^2)(1 - v^2)}{(1 + uv)^2} > 0 \quad (|u|, |v| < 1).$$

Only the first two child signals below will require the sharper loss for means bounded by θ .

3.2 The amount of information in one component

For a lower bound, we use a weighted linear estimator in the tradition of second-moment reconstruction estimates on trees [6]; see also Mossel–Peres [13, Section 5, Lemma 5.3]. Here it uses the Gaussian observations at all depths up to a cutoff. The cutoff balances the independent noise against the correlations among the spins. For the upper bound, the fusion loss gives the complementary estimate.

Lemma 3.2 (Critical posterior information). *There are constants $c, C > 0$, depending only on d , such that for every $t > 0$ and every vertex v ,*

$$c\omega(t) \leq M(t) \leq C\omega(t), \quad \mathbb{E}m_v(t)^2 \leq C\omega(t). \quad (3.2)$$

One may take $c = 1/5$. In particular both upper bounds are $C\sqrt{t}$ for $0 < t \leq 1$.

Proof. Root $T_{v \rightarrow w}$ at v . It is a rooted b -ary tree; write $|u|$ for the depth of u in it. For an integer $n \geq 0$, form the component-observable statistic

$$Z_n = \sum_{|u| \leq n} \theta^{|u|} X_u(t).$$

The broadcast covariance is $\mathbb{E}[\sigma_u \sigma_z] = \theta^{\text{dist}(u, z)}$. Since $b\theta^2 = 1$,

$$\mathbb{E}[\sigma_v Z_n] = t \sum_{k=0}^n b^k \theta^{2k} = t(n + 1). \quad (3.3)$$

The Brownian part of Z_n has variance $t(n + 1)$ and is independent of its spin part.

We count the latter part's second moment. If an ordered pair (u, z) has depths k, ℓ and its last common ancestor has depth j , its weighted spin covariance, before the factor t^2 , is

$$\theta^{k+\ell+\text{dist}(u,z)} = \theta^{2(k+\ell-j)} = b^{-(k+\ell-j)}.$$

When $j < \min(k, \ell)$, choose the common ancestor, the two distinct first children in their ordered roles, and then the remaining descendants. The exact count is

$$b^j b(b-1) b^{k-j-1} b^{\ell-j-1} = (b-1) b^{k+\ell-j-1}.$$

When $j = \min(k, \ell)$, the shallower vertex is an ancestor of the deeper one. There are $b^{\max(k, \ell)}$ such ordered pairs; this includes the b^k diagonal pairs when $k = \ell = j$. Each count is at most $b^{k+\ell-j}$. Thus the weighted contribution for each triple (k, ℓ, j) is at most one, and

$$\begin{aligned} \mathbb{E} Z_n^2 &\leq t(n+1) + t^2 \sum_{k, \ell=0}^n (\min(k, \ell) + 1) \\ &= t(n+1) + t^2 \sum_{j=0}^n (n-j+1)^2 \leq t(n+1) + t^2(n+1)^3. \end{aligned} \quad (3.4)$$

Because Z_n is measurable from the component observations, $\mathbb{E}[\sigma_v Z_n] = \mathbb{E}[(\tanh Y) Z_n]$. Cauchy-Schwarz and (3.3)–(3.4) therefore give

$$M(t) \geq \frac{t(n+1)}{1+t(n+1)^2}. \quad (3.5)$$

For $0 < t \leq 1$, take $n = \lfloor t^{-1/2} \rfloor$. Then $t^{-1/2} \leq n+1 \leq 2t^{-1/2}$, so the last bound is at least $\sqrt{t}/5$. For $t \geq 1$, take $n = 0$ to obtain $M(t) \geq t/(1+t) \geq 1/2$. This proves the lower bound with $c = 1/5$.

For the upper bound, condition on the component root spin. Its own observation history and the b child-component histories are conditionally independent. The own signal has posterior mean $\tanh X_v(t)$, by the one-spin version of (2.2), and

$$q_0(t) := \mathbb{E} \tanh^2 X_v(t) \leq \mathbb{E} X_v(t)^2 = t + t^2.$$

One child's entire observation history has posterior mean $\theta \tanh Y$ about the root spin: conditional on the child spin, the root has mean θ times that spin. Its second moment is $\theta^2 M(t)$, and its magnitude is at most θ . These statements concern full infinite component histories; their conditional independence follows from the broadcast construction and the independent Brownian noises, first for cylinder events and then for the sigma-fields they generate.

There are at least two child signals because $b \geq 2$. Combine two of them first. Lemma 3.1 subtracts at least $2(1 - \theta^2) \theta^4 M(t)^2$. Add the other $b - 2$ child signals and the own signal by subadditivity. The combined signal is the full component history, and hence

$$M(t) \leq b \theta^2 M(t) - 2(1 - \theta^2) \theta^4 M(t)^2 + q_0(t).$$

The linear terms cancel at $b \theta^2 = 1$. Since $q_0(t) \leq 2t$ for $t \leq 1$, this yields $M(t) \leq C \sqrt{t}$. At a whole vertex there are d child components, so subadditivity alone gives

$$\mathbb{E} m_v(t)^2 \leq q_0(t) + d \theta^2 M(t) \leq C \sqrt{t} \quad (0 < t \leq 1).$$

Both second moments are at most one for $t \geq 1$, completing (3.2). $\square$

We have now identified the critical information scale. To turn its lower bound into decay along a path, we need the conditional variation of a side branch's outgoing message, even when the attaching spin has been specified.

Lemma 3.3 (Conditioning a component on a neighboring spin). *For every integrable function $f(Y)$ and $\eta \in \{-1, 1\}$,*

$$\begin{aligned}\mathbb{E}[f(Y) \mid \sigma_v = \eta] &= \mathbb{E}[f(Y)(1 + \eta \tanh Y)], \\ \mathbb{E}[f(Y) \mid \sigma_w = \eta] &= \mathbb{E}[f(Y)(1 + \eta \theta \tanh Y)].\end{aligned}\tag{3.6}$$

In particular either conditioning leaves the expectation of every integrable even function of Y unchanged.

Proof. Let $\mathcal{G} = \mathcal{F}_t^{T_{v \rightarrow w}}$. Since σ_v is a uniform sign and $\mathbb{E}[\sigma_v \mid \mathcal{G}] = \tanh Y$, conditioning the indicator of $\{\sigma_v = \eta\}$ on $\mathcal{G}$ proves the first identity. The component observations are independent of σ_w conditional on σ_v , and $\mathbb{E}[\sigma_w \mid \sigma_v] = \theta \sigma_v$. It follows that $\mathbb{E}[\sigma_w \mid \mathcal{G}] = \theta \tanh Y$; the same indicator calculation proves the second identity. The multiplying densities are bounded, so the formulas apply to every marginally integrable $f(Y)$. If f is even, its product with $\tanh Y$ is odd. Symmetry of the marginal law of Y makes that correction vanish. $\square$

3.3 Attenuation from an intact side branch

The next argument uses the full infinite component beyond one side edge. Its conditioning specifies the spins on a deterministic path, while leaving the selected side branches unobserved. This is the setting in which the information lower bound supplies a loss uniformly over the contribution from the rest of the tree.

Lemma 3.4 (Conditional path attenuation). *There is $c_0 > 0$, depending only on d , such that for every $t > 0$, every $z \in \mathbb{R}$, and $\eta \in \{-1, 1\}$,*

$$\mathbb{E} \left[a(z + \psi(h_{v \rightarrow w}(t)))^2 \mid \sigma_w = \eta \right] \leq 1 - c_0 \omega(t).\tag{3.7}$$

For every deterministic simple path $v_0, \dots, v_r$, $r \geq 1$,

$$\mathbb{E} \left[\prod_{i=1}^r a(h_{v_i \rightarrow v_{i-1}}(t))^2 \mid \sigma_{v_0}, \dots, \sigma_{v_r} \right] \leq e^{-c_0 \omega(t)r}.\tag{3.8}$$

The same bound holds for the reverse product $\prod_{i=1}^r a(h_{v_{i-1} \rightarrow v_i}(t))^2$, and both bounds also hold without conditioning on the path spins.

Proof. Set $G = \psi(Y)$ and condition on the attaching spin $\sigma_w = \eta$. We have $|G| \leq \beta$. The second identity in (3.6), symmetry, and Cauchy–Schwarz give

$$\mathbb{E}[G^2 \mid \sigma_w = \eta] = \mathbb{E}G^2, \quad |\mathbb{E}[G \mid \sigma_w = \eta]|^2 = \theta^2 |\mathbb{E}[G \tanh Y]|^2 \leq \theta^2 \mathbb{E}G^2.$$

Moreover $|\psi(y)| \geq \theta |\tanh y|$, since $|\tanh u| \geq |u|$. Consequently

$$\text{Var}(G \mid \sigma_w = \eta) \geq (1 - \theta^2) \mathbb{E}G^2 \geq (1 - \theta^2) \theta^2 M(t).\tag{3.9}$$

The conditional variance is the useful quantity here: unlike an uncentered second moment, it cannot be eliminated by adding a shift.

For any random variable G supported on $[-\beta, \beta]$, there is a constant $c_\beta > 0$ such that

$$\mathbb{E} \tanh^2(z + G) \geq c_\beta \text{Var}(G) \quad \text{for every } z \in \mathbb{R}.\tag{3.10}$$

To see this, first suppose $|z| \leq \beta + 1$. Then $|z + G| \leq 2\beta + 1$. The ratio $\tanh^2 u / u^2$, extended continuously to be one at $u = 0$, has a positive minimum on that compact interval. Hence $\mathbb{E} \tanh^2(z + G) \geq c_\beta \mathbb{E}(z + G)^2 \geq c_\beta \text{Var}(G)$. If $|z| > \beta + 1$, then $|z + G| > 1$, so the left side is at least $\tanh^2(1)$; use $\text{Var}(G) \leq \beta^2$ and decrease c_β if needed. This proves (3.10) for the conditional laws in (3.9) as well.

For $u = \tanh^2 h$, the elementary bounds on a give

$$a(h)^2 \leq a(h) = 1 - \frac{(1 - \theta^2)u}{1 - \theta^2 u} \leq 1 - (1 - \theta^2)u.$$

Apply this with $h = z + G$. Equations (3.10) and (3.9), followed by $M(t) \geq \omega(t)/5$, prove (3.7), for example with

$$c_0 = \min \left\{ \frac{1}{2}, \frac{c_\beta(1 - \theta^2)^2 \theta^2}{5} \right\}.$$

The same estimate may be used with a random shift after conditioning on data that fix the shift and are independent of the selected branch given its attaching spin.

For the path estimate, choose at each target v_i , using only the path geometry, one neighbor s_i that is not on the path. There are $d - 2 \geq 1$ choices at an internal target and $d - 1$ at the last target. The components $R_i = T_{s_i \rightarrow v_i}$ are disjoint and infinite. In particular, when $d = 3$ an internal target still has one full infinite side component.

Let

$$\mathcal{H} = \sigma(\sigma_{v_0}, \dots, \sigma_{v_r}, X_u(s) : u \notin \bigcup_i R_i, 0 \leq s \leq t).$$

Conditional on $\mathcal{H}$, the selected component observation arrays are independent, and the law in R_i depends only on its attaching spin σ_{v_i} . This follows directly by rooting the broadcast at v_0 : the multiplier on each attaching edge, the multipliers inside its selected component, and its Brownian motions form one independent group. The outside observations use none of these groups once the path spins are specified. The assertion for the full observation arrays follows from the corresponding assertion on finite cylinder events.

At this fixed time, the cavity identity gives, almost surely,

$$h_{v_i \rightarrow v_{i-1}} = \psi(h_{s_i \rightarrow v_i}) + \left(X_{v_i} + \sum_{\substack{u \sim v_i \\ u \neq v_{i-1}, s_i}} \psi(h_{u \rightarrow v_i}) \right).$$

When all other selected arrays are held fixed, the parenthesized quantity is a fixed shift independent of the array in R_i conditional on σ_{v_i} . Integrate R_1 first. Every factor at a later target v_j , $j > 1$, uses its component on the side excluding v_{j-1} , and hence does not use R_1 . Thus (3.7) removes the first squared factor at a cost at most $1 - c_0 \omega(t)$. Repeating in the order $R_2, \dots, R_r$ gives

$$\mathbb{E} \left[\prod_{i=1}^r a(h_{v_i \rightarrow v_{i-1}})^2 \middle| \mathcal{H} \right] \leq (1 - c_0 \omega(t))^r \leq e^{-c_0 \omega(t)r}.$$

Taking a conditional expectation given only the path spins proves (3.8). Reversing the deterministic path gives the stated reverse product. Finally average over its spins. $\square$

The loss above comes from the posterior information in an intact infinite branch. It is not a claim about arbitrarily truncated branches: a finite boundary will require a separate estimate using the target's own Gaussian observation. For the present response row, the infinite-branch estimate is exactly what is needed.

3.4 Time-integrable response rows

Return to the full range $0 < \theta \leq b^{-1/2}$. For the progressive coefficients in (2.6), define the extended nonnegative sum

$$A_v(t) = \sum_{u \in \mathcal{V}} c_{vu}(t)^2.$$

It is measurable as a countable sum. The following estimate is the input that permits division by the posterior variance when the filtering equation is converted to a field equation.

Proposition 3.5 (Time-integrable response rows). *For every vertex v , there is $C < \infty$, depending only on d at the critical parameter, such that*

$$\begin{aligned} A_v(t) &\leq 1 + \frac{d\theta^2}{1 - b\theta^2} \quad \text{if } b\theta^2 < 1, \quad t \geq 0, \\ \mathbb{E}A_v(t) &\leq \frac{C}{\omega(t)} \quad \text{if } b\theta^2 = 1, \quad t > 0, \\ \mathbb{E} \int_0^T A_v(t) dt &< \infty \quad \text{for every } T < \infty \text{ in either case.} \end{aligned} \tag{3.11}$$

The same fixed-time and integrated upper bounds hold with $\sum_u C_{vu}(t)^2$ in place of $A_v(t)$.

Proof. There are db^{r-1} vertices at distance $r \geq 1$ from v . If $b\theta^2 < 1$, the pointwise bound $a \leq 1$ gives

$$A_v(t) \leq 1 + \sum_{r \geq 1} db^{r-1} \theta^{2r} = 1 + \frac{d\theta^2}{1 - b\theta^2}.$$

At equality, $db^{r-1}\theta^{2r} = d/b$. Tonelli's theorem and Lemma 3.4 give, for $t > 0$,

$$\mathbb{E}A_v(t) \leq 1 + \frac{d}{b} \sum_{r \geq 1} e^{-c_0\omega(t)r} = 1 + \frac{d/b}{e^{c_0\omega(t)} - 1} \leq 1 + \frac{d}{bc_0\omega(t)} \leq \frac{C}{\omega(t)}.$$

Here $e^x - 1 \geq x$ and $\omega(t) \leq 1$. Since

$$\int_0^T \frac{dt}{\omega(t)} = \begin{cases} 2\sqrt{T}, & 0 \leq T \leq 1, \\ T + 1, & T \geq 1, \end{cases}$$

another application of Tonelli proves the integrated bound. Finally $\sum_u C_{vu}^2 = (1 - m_v^2)^2 A_v \leq A_v$ by (2.6). $\square$

In particular, $A_v(t)$ is finite almost surely at each fixed positive time and is finite for almost every time almost surely on each compact horizon. At criticality the raw sum at $t = 0$ is infinite: all fields vanish, $a(0) = 1$, and every positive-distance shell contributes d/b . That single time has zero Lebesgue measure. The integrated estimate, rather than a bound at time zero, is the conclusion used in the construction of the infinite innovation equation.

4 Constructing the infinite innovation equation

The posterior fields describe how the observations affect the spins. We now turn that description into a stochastic equation driven by independent Brownian motions. This is the equation that will allow us to compare two observation processes driven by the same noise. Throughout the section,

$$b = d - 1, \quad 0 < \theta = \tanh \beta \leq b^{-1/2}.$$

We use the raw observation versions from Lemma 2.1. In particular, at this stage a posterior identity holds almost surely at each fixed time, and therefore outside a $dt d\mathbb{P}$ -null set; continuity of the infinite filter has not yet been established.

For each vertex define its *innovation* by

$$W_v(t) = X_v(t) - \int_0^t m_v(s) ds. \quad (4.1)$$

The integrand is a bounded, progressively measurable observation function. Thus this formula defines continuous paths for every observation input. Their values through time t use only the observation history through t . The time integrals are Borel functions of that history, so $X \mapsto W$ is a causal Borel map on the countable product of continuous path spaces. The centered-ball definitions of the posterior means also show that this map commutes with graph relabeling.

The innovation construction is classical in nonlinear filtering [7] and is the starting point of the Brownian Ising construction of Nam, Sly, and Zhang [14]. Here we must justify its equation for countably many observations. We will first prove that W is Brownian, then identify the entire posterior martingale using an exact energy limit. Neither step assumes that W determines X .

Proposition 4.1 (Infinite innovation equations). *In the planted observation law $X_v(t) = t\sigma_v + B_v(t)$, every finite subvector of $(W_v)_{v \in \mathcal{V}}$ is a standard vector Brownian motion in both the raw observation filtration $(\mathcal{F}_t^0)$ and its usual augmentation $(\mathcal{F}_t)$. In particular W has product Wiener law.*

The posterior means have continuous versions, starting at zero, and

$$m_v(t) = \sum_{u \in \mathcal{V}} \int_0^t C_{vu}(s) dW_u(s). \quad (4.2)$$

At each fixed time these versions are $\mathbb{E}[\sigma_v | \mathcal{F}_t]$, and C_{vu} is the covariance conditional on $\mathcal{F}_t$. The site fields and the component fields can be chosen as jointly Borel, progressively measurable functions of time and observation history that are continuous in time almost surely. For every input, the chosen function $h_{v \rightarrow w}(t)$ uses only observations in $T_{v \rightarrow w}$ through time t . Almost surely the cavity identities and the bounds (2.5) then hold simultaneously at all times and all their indices.

With the response coefficients c_{vu} from (2.6) and $A_v = \sum_u c_{vu}^2$ from (3.11), the continuous site field satisfies the equation below. In every integral, A_v is assigned value zero wherever this sum is infinite.

$$h_v(t) = \sum_{u \in \mathcal{V}} \int_0^t c_{vu}(s) dW_u(s) + \int_0^t m_v(s) A_v(s) ds. \quad (4.3)$$

Each Brownian sum in this proposition converges, along any exhaustion by finite vertex sets, in the norm

$$\|Z\|_{\mathcal{M}_T^2} = \left(\mathbb{E} \sup_{0 \leq t \leq T} |Z(t)|^2 \right)^{1/2} \quad (T < \infty).$$

The drift in (4.3) is absolutely integrable on each such horizon almost surely and in expectation.

The raw and augmented filtrations both occur in the proof. We record the elementary passage between them once. If $(\mathcal{G}_t^0)$ is a raw filtration, write $\mathcal{G}_t = \bigcap_{r>t} \overline{\mathcal{G}_r^0}$ for its usual augmentation, where the bar includes the null sets of the ambient probability space.

Lemma 4.2 (Passage to the usual augmentation). *If Y is a continuous square-integrable martingale in $(\mathcal{G}_t^0)$ and is continuous in L^2 at each time, then it is also a martingale in $(\mathcal{G}_t)$. More generally, if Z is integrable and a version $M(t) = \mathbb{E}[Z | \mathcal{G}_t^0]$ is continuous in L^1 , then $M(t) = \mathbb{E}[Z | \mathcal{G}_t]$ at every fixed time.*

Proof. For $s < t$, choose $s_n \downarrow s$ with $s_n < t$. The completed sigma-fields $\overline{\mathcal{G}_{s_n}^0}$ decrease to $\mathcal{G}_s$. Reverse conditional expectation convergence gives

$$\mathbb{E}[Y(t) \mid \mathcal{G}_s] = \lim_n \mathbb{E}[Y(t) \mid \overline{\mathcal{G}_{s_n}^0}] = \lim_n Y(s_n) = Y(s)$$

in L^1 ; the last limit even holds in L^2 . Completion does not change conditional expectations up to null sets. The same argument, with Z in place of $Y(t)$, proves the second assertion. A continuous bounded process is continuous in every finite L^p , by dominated convergence, so the second assertion applies to the bounded posterior versions constructed below. $\square$

4.1 Brownian innovations and finite posterior energy

We first prove the asserted Brownian property. For $0 \leq s < t$, the future increments of the planted Brownian motions are independent of all spins and of their Brownian pasts. Conditioning on the smaller raw observation history therefore gives

$$\mathbb{E}[X_v(t) - X_v(s) \mid \mathcal{F}_s^0] = (t - s)m_v(s).$$

For every deterministic $r \geq s$, the tower property for the bounded raw posteriors also gives $\mathbb{E}[m_v(r) \mid \mathcal{F}_s^0] = m_v(s)$. Conditional Fubini is applicable because $|m_v| \leq 1$. Subtracting the conditional means of the two increments in (4.1) shows that W_v is a continuous square-integrable martingale in $(\mathcal{F}_t^0)$.

This conclusion concerns the innovations, not the planted noises. In the filtration containing all spins at time zero and all Brownian histories, their relation is

$$W_v(t) = B_v(t) + \int_0^t (\sigma_v - m_v(s)) \, ds.$$

The added path has finite variation. Hence the pathwise quadratic covariations are

$$[W_v, W_u]_t = \mathbf{1}_{\{v=u\}} t.$$

The bounded drift and L^2 continuity of B_v give L^2 continuity of W_v . Lemma 4.2 makes each coordinate a martingale in $(\mathcal{F}_t)$. Lévy's characterization applied to each finite vector, with the displayed covariations, proves that vector is standard Brownian motion in $(\mathcal{F}_t)$ and hence also in the smaller raw filtration. Its future increments are independent of the observation past. By a monotone-class argument over finite vertex sets and rational future times, the entire future increment field $(W_v(t+s) - W_v(t))_{v \in \mathcal{V}, s \geq 0}$ is independent of $\mathcal{F}_t$ for every fixed t .

Finite observation sets have their own innovations. They will provide an exact scalar energy identity, but we will never identify their Brownian paths with the infinite innovations.

Lemma 4.3 (Finite posterior energy). *Let D be a finite connected subtree. For $u \in D$ put*

$$W_u^D(t) = X_u(t) - \int_0^t m_u^D(s) \, ds.$$

The vector W^D is standard Brownian motion in the usual augmentation of $(\mathcal{F}_t^D)$, and, for $v \in D$,

$$\begin{aligned} m_v^D(t) &= \sum_{u \in D} \int_0^t C_{vu}^D(s) \, dW_u^D(s), \\ \mathbb{E}[m_v^D(t)^2] &= \mathbb{E} \int_0^t \sum_{u \in D} C_{vu}^D(s)^2 \, ds. \end{aligned} \tag{4.4}$$

Proof. The preceding conditional-increment and covariation calculation works with the observations restricted to D , and proves the Brownian assertion. The finite posterior formula (2.2) gives $m_v^D(t) = M_v^D(X_D(t))$ for a smooth function M_v^D . It is a bounded continuous martingale in the raw filtration, since it is a version of $\mathbb{E}[\sigma_v \mid \mathcal{F}_t^D]$. It is continuous in L^2 , so Lemma 4.2 also makes it a martingale in the augmented finite observation filtration.

Apply finite-dimensional Itô's formula to $M_v^D(X_D)$ using $dX_u = m_u^D dt + dW_u^D$. By (2.3), its continuous local martingale part is $\sum_{u \in D} \int C_{vu}^D dW_u^D$. The remaining part is continuous and has finite variation locally. Since m_v^D is itself a martingale, that part is also a local martingale and must vanish. The covariance coefficients are bounded, so the finite Brownian sum is square integrable on every finite horizon. The free prior has mean zero. This proves the first identity in (4.4), and orthogonality of the Brownian coordinates and Itô's isometry prove the second. $\square$

4.2 No covariance energy escapes to the boundary

The finite energy identity does not yet imply its infinite analogue: coordinatewise convergence could leave a positive amount of energy at ever more distant vertices. We control that possibility in two intervals. Away from time zero, each vertex's own Gaussian noise attenuates its path factor, including at a finite-tree leaf. Near time zero, the whole-site information estimate bounds the entire finite energy at once.

Lemma 4.4 (Gaussian attenuation away from time zero). *For every $\delta > 0$ there is $\kappa_\delta \in (0, 1)$ such that, for every finite connected subtree D , every $v, u \in D$, and every $s \geq \delta$,*

$$\mathbb{E}[C_{vu}^D(s)^2] \leq \theta^{2r} (1 - \kappa_\delta)^r, \quad r = \text{dist}(v, u). \quad (4.5)$$

The same inequality holds for the whole-tree covariance $C_{vu}(s)$.

Proof. Choose $\varepsilon_\delta > 0$ with $2\varepsilon_\delta/\sqrt{2\pi\delta} \leq 1/2$. A Gaussian random variable Z of variance $s \geq \delta$, with any mean, has $\mathbb{P}(|Z| \leq \varepsilon_\delta) \leq 1/2$ by its density bound. Since

$$a(z)^2 \leq a(z) \leq 1 - (1 - \theta^2) \tanh^2 z,$$

it follows that

$$\mathbb{E}[a(Z)^2] \leq 1 - \kappa_\delta, \quad \kappa_\delta = \frac{1 - \theta^2}{2} \tanh^2 \varepsilon_\delta \in (0, 1).$$

The estimate is uniform in the mean.

Let $v = v_0, \dots, v_r = u$ be the path in D . In the squared covariance formula, discard the initial factor $(1 - (m_v^D)^2)^2 \leq 1$. At the fixed time s the remaining factors depend only on terminal observations. Conditional on all spins and all terminal Brownian coordinates other than $B_{v_1}(s)$, the cavity recursion has the form

$$h_{v_1 \rightarrow v_0}^D(s) = B_{v_1}(s) + z,$$

where the shift z is fixed: all branch fields in this recursion exclude v_1 . The later fields $h_{v_i \rightarrow v_{i-1}}^D$, $i > 1$, also exclude v_1 . Conditional on the spins, the terminal Brownian coordinates are independent Gaussians of variance s . Integrating $B_{v_1}(s)$ first therefore removes the first squared factor at a cost at most $1 - \kappa_\delta$, without changing the later ones. Integrate $B_{v_2}(s), \dots, B_{v_r}(s)$ in that order. Iteration proves (4.5), even conditional on the spins. At a leaf the shift is simply its planted spin drift; the argument still applies. No infinite side branch is used here.

For fixed v, u, s , exhaust the tree by finite connected subtrees containing the path. Bounded conditional-expectation convergence from Lemma 2.1 gives $C_{vu}^D(s) \rightarrow C_{vu}(s)$ in L^2 . Passing to the limit proves the whole-tree inequality. $\square$

Fix v and let D_R be its radius- R ball. Extend $C_{vu}^{D_R}$ by zero for $u \notin D_R$. For each fixed u and $T < \infty$,

$$C_{vu}^{D_R} \longrightarrow C_{vu} \quad \text{in } L^2([0, T] \times \Omega, ds \, d\mathbb{P}). \quad (4.6)$$

Indeed, at each deterministic s , increasing-sigma-field convergence applies to the bounded variables σ_v , σ_u , and $\sigma_v \sigma_u$. Their conditional means determine the covariance. The covariances are uniformly bounded, so dominated convergence also integrates the resulting L^2 convergence in time.

When $b\theta^2 < 1$, the pointwise bound $C_{vu}^{D_R}(s)^2 \leq \theta^{2 \operatorname{dist}(v,u)}$ controls all spatial tails: there are $n_r = db^{r-1}$ vertices at distance $r \geq 1$, and

$$\sup_R \mathbb{E} \int_0^T \sum_{\operatorname{dist}(v,u) > L} C_{vu}^{D_R}(s)^2 ds \leq \frac{d}{b} T \sum_{r > L} (b\theta^2)^r \longrightarrow 0.$$

The same deterministic estimate holds for the infinite coefficients.

At equality, fix $0 < \delta \leq \min\{T, 1\}$. The tower property gives $m_v^{D_R}(\delta) = \mathbb{E}[m_v(\delta) \mid \mathcal{F}_\delta^{D_R}]$. Conditional Jensen, the finite energy identity, and the whole-site estimate in (3.2) yield

$$\mathbb{E} \int_0^\delta \sum_u C_{vu}^{D_R}(s)^2 ds = \mathbb{E}[m_v^{D_R}(\delta)^2] \leq \mathbb{E}[m_v(\delta)^2] \leq C\sqrt{\delta}. \quad (4.7)$$

On $[\delta, T]$, Lemma 4.4 applies uniformly in R , and now $n_r \theta^{2r} = d/b$. Thus

$$\sup_R \mathbb{E} \int_0^T \sum_{\operatorname{dist}(v,u) > L} C_{vu}^{D_R}(s)^2 ds \leq C\sqrt{\delta} + \frac{d}{b} T \sum_{r > L} (1 - \kappa_\delta)^r. \quad (4.8)$$

This also bounds the infinite tail. To see this without assuming its finiteness, first restrict that tail to a finite annulus, pass $R \rightarrow \infty$ using (4.6), and then increase the outer radius using monotone convergence. First send $L \rightarrow \infty$ with δ fixed, so that the geometric term vanishes. Then send $\delta \downarrow 0$. The order matters: κ_δ may shrink as δ tends to zero. We have proved that the integrated spatial tails of both the finite and the infinite rows vanish uniformly.

For a fixed L , the sum over $\operatorname{dist}(v, u) \leq L$ is finite. Equation (4.6) passes its squared energy to the limit in L^1 . The uniform tail estimates then pass the complete energy to the limit. At each fixed t we also have $m_v^{D_R}(t) \rightarrow m_v(t)$ in L^2 . Consequently (4.4) gives the exact infinite identity

$$\mathbb{E}[m_v(t)^2] = \mathbb{E} \int_0^t \sum_{u \in \mathcal{V}} C_{vu}(s)^2 ds \quad (t \geq 0). \quad (4.9)$$

At $t = 0$ both sides are zero. Only the scalar energies have been passed to the limit; the distinct processes W^{D_R} have played no part in this limiting operation.

4.3 Identifying the entire posterior martingale

The energy identity gives the norm that a Brownian representation must have. A second identity identifies its inner product with the posterior. Together they determine the posterior itself and exclude any additional martingale orthogonal to the innovations.

We first record the approximation that permits coefficients to be used in this test. It will also let the subsequent conditional-copy construction transport individual stochastic integrals.

Lemma 4.5 (Observation-adapted step approximation). *For fixed v, u and $T < \infty$, each of C_{vu} and c_{vu} is the $L^2([0, T] \times \Omega, ds \, d\mathbb{P})$ limit of bounded elementary processes*

$$G(s) = \sum_{j=0}^{N-1} \xi_j \mathbf{1}_{(t_j, t_{j+1}]}(s), \quad 0 = t_0 < \dots < t_N = T,$$

where ξ_j is a bounded Borel function of finitely many observations at time t_j . In particular each coefficient has a predictable representative for stochastic integration.

Proof. For R large enough to contain the path from v to u , the finite coefficient $C_{vu}^{D_R}$ is a bounded continuous function of the terminal observations in D_R , and is therefore a continuous raw adapted process. It converges to C_{vu} in the required space by (4.6). For the response coefficient use

$$c_{vu}^{D_R} = \theta^{\text{dist}(v,u)} \prod_{i=1}^{\text{dist}(v,u)} a(h_{v_i \rightarrow v_{i-1}}^{D_R}).$$

The component convergence in Lemma 2.1 gives fixed-time convergence to c_{vu} ; the common bound $|c_{vu}^{D_R}| \leq \theta^{\text{dist}(v,u)} \leq 1$ again gives convergence in the stated L^2 space. This finite coefficient is also a continuous raw adapted function of finitely many terminal observations.

For either finite coefficient, sample its values at the left endpoints of a deterministic mesh. As the mesh tends to zero, continuity and the common bound imply convergence in $L^2(ds \, d\mathbb{P})$ by dominated convergence. Choose a diagonal sequence of radii and meshes. These are the asserted elementary processes. A subsequence converges $ds \, d\mathbb{P}$ -almost everywhere; its pointwise limit where finite, with value zero elsewhere, is predictable because each elementary process is predictable. This representative agrees with the original coefficient in the required product-measure sense. $\square$

Fix v, u and $t < \infty$. For any bounded elementary process of the same form whose ξ_j is merely $\mathcal{F}_{t_j}^0$ -measurable, we claim

$$\mathbb{E} \left[m_v(t) \int_0^t G(s) \, dW_u(s) \right] = \mathbb{E} \int_0^t G(s) C_{vu}(s) \, ds. \quad (4.10)$$

For an elementary process, the integral is the finite sum of $\xi_j(W_u(t_{j+1}) - W_u(t_j))$, hence is measurable in $\mathcal{F}_t^0$. In its expectation against $m_v(t)$ we may therefore replace $m_v(t)$ by σ_v . In the planted filtration,

$$dW_u(s) = dB_u(s) + (\sigma_u - m_u(s)) \, ds.$$

Here $\sigma_v G$ is a bounded predictable integrand: σ_v is known at time zero and every raw observation past is contained in the planted past. Thus $\mathbb{E}[\sigma_v \int_0^t G \, dB_u] = 0$. For the finite-variation part, condition at time s on the raw observation history:

$$\mathbb{E}[\sigma_v(\sigma_u - m_u(s)) \mid \mathcal{F}_s^0] = \mathbb{E}[\sigma_v \sigma_u \mid \mathcal{F}_s^0] - m_v(s) m_u(s) = C_{vu}(s).$$

Boundedness permits Fubini, proving (4.10). This argument has used only the fact that W is a function of the observation past, not any converse measurability.

Apply Lemma 4.5 to C_{vu} . Itô's isometry for W_u and Cauchy-Schwarz on the two sides of (4.10) pass to the limit, so the identity also holds with $G = C_{vu}$. For a finite vertex set F let

$$S_{v,F}(t) = \sum_{u \in F} \int_0^t C_{vu}(s) \, dW_u(s).$$

The integrals for different u are orthogonal, even though their integrands may depend on every observation: their driving Brownian coordinates have zero mutual quadratic covariation. The isometry and the testing identity give

$$\mathbb{E}[S_{v,F}(t)^2] = \sum_{u \in F} \mathbb{E} \int_0^t C_{vu}(s)^2 ds = \mathbb{E}[m_v(t)S_{v,F}(t)].$$

Combining this with the exact energy (4.9) yields the useful squared-error identity

$$\mathbb{E}[(m_v(t) - S_{v,F}(t))^2] = \sum_{u \notin F} \mathbb{E} \int_0^t C_{vu}(s)^2 ds. \quad (4.11)$$

The right side tends to zero as F exhausts the vertices. This is precisely the step that eliminates a possible residual martingale.

There is also convergence of the processes, rather than just their values at a fixed time. For $F \subseteq F'$, Doob's maximal L^2 inequality gives

$$\mathbb{E} \sup_{0 \leq s \leq T} |S_{v,F'}(s) - S_{v,F}(s)|^2 \leq 4 \sum_{u \in F' \setminus F} \mathbb{E} \int_0^T C_{vu}(s)^2 ds.$$

The total energy is finite by (4.9). The partial sums therefore converge in $\mathcal{M}_T^2$ to a continuous square-integrable martingale. The limits agree on overlapping horizons, and (4.11) identifies each fixed-time value with the raw posterior $m_v(t)$. Countability of the vertices lets us choose these continuous versions simultaneously. This proves (4.2).

4.4 Continuous versions that retain component locality

The Brownian representation has supplied a continuous version of each site mean. We now select Borel observation functions realizing this version and continuous component fields. Constructing component fields from site fields could apparently use information from the opposite side of an edge. We first establish continuous component modifications, then choose them by explicit past-observation formulas that retain the required component restriction.

Write M_v for the continuous mean version temporarily. At every rational time it agrees almost surely with the raw mean from Lemma 2.1. Intersect these events over all vertices and rational times. The raw field bound and continuity of M_v and X_v imply, on this one event,

$$|M_v(t)| \leq \tanh(|X_v(t)| + d\beta) < 1 \quad (t \geq 0).$$

Indeed the inequality holds first at rational times and then passes to every time by continuity. On each compact interval,

$$\sup_{0 \leq t \leq T} |M_v(t)| \leq \tanh \left(\sup_{0 \leq t \leq T} |X_v(t)| + d\beta \right) < 1.$$

Thus $H_v = \operatorname{atanh} M_v$ is continuous. The stronger inequality $|H_v - X_v| \leq d\beta$ also passes from rational times to all times.

For adjacent vertices, consider for arbitrary real H, K the equations

$$x + \psi(y) = H, \quad y + \psi(x) = K.$$

They have a unique solution. In fact x is the unique fixed point of

$$x \mapsto H - \psi(K - \psi(x)),$$

a contraction of the complete space $\mathbb{R}$ with constant at most $\theta^2 < 1$, and $y = K - \psi(x)$. Comparing two solutions gives

$$|x - \tilde{x}| \leq \frac{|H - \tilde{H}| + \theta|K - \tilde{K}|}{1 - \theta^2},$$

and the analogous bound for y . The solution therefore depends continuously on (H, K) .

Apply this construction to $(H_v(t), H_w(t))$ and denote the resulting continuous processes by $(H_{v \rightarrow w}^c, H_{w \rightarrow v}^c)$. At every fixed time the raw cavity identities identify this solution almost surely with the raw component fields, by uniqueness. After intersecting over rational times and oriented edges, continuity therefore extends both cavity identities and the component bound to all times. This proves existence of continuous component modifications. The construction by adjacent site fields is used only for that existence assertion.

We now specify the observation functions actually used in the rest of the proof. Let $m_v^0(t, x)$ and $h_{v \rightarrow w}^0(t, x)$ be the original raw functions obtained from the centered finite-ball limits in Lemma 2.1. For $q_n(t) = 2^{-n} \lfloor 2^n t \rfloor$, set

$$\hat{m}_v(t, x) = \begin{cases} \lim_{n \rightarrow \infty} m_v^0(q_n(t), x), & \text{if the limit exists in } (-1, 1), \\ 0, & \text{otherwise,} \end{cases}$$

and set $\hat{h}_v = \text{atanh } \hat{m}_v$. For every oriented edge, set

$$\hat{h}_{v \rightarrow w}(t, x) = \begin{cases} \lim_{n \rightarrow \infty} h_{v \rightarrow w}^0(q_n(t), x), & \text{if the limit exists in } \mathbb{R}, \\ 0, & \text{otherwise.} \end{cases}$$

For each n the sampled process is constant on the half-open dyadic time intervals, and its value there is a Borel function of the appropriate observation history at the left endpoint. It is therefore progressively measurable. Real convergence is a countable Cauchy condition. For the site selector, membership of the resulting limit in $(-1, 1)$ is an additional Borel condition. Taking these limits with the displayed defaults therefore preserves Borel and progressive measurability. For the component expression both the entire sequence and its default decision use only observations in $T_{v \rightarrow w}$ through time t . In particular, two inputs agreeing on that component history have the same sequence and the same chosen value. This is pointwise component locality, including on inputs where the limit fails.

At every dyadic time the raw site mean agrees almost surely with M_v , and the raw component field agrees almost surely with $H_{v \rightarrow w}^c$. Intersecting over the countable dyadic times and indices gives one event of probability one. On that event, for every real t the sampled values above are values of the respective continuous modifications at $q_n(t)$. Since $q_n(t) \rightarrow t$, continuity makes their limits agree simultaneously for every t . Thus the chosen functions are indistinguishable from the continuous modifications. They retain all the all-time cavity identities and bounds just proved. The same dyadic rule and the same defaults at every index also preserve the graph-relabeling covariance of the raw functions.

We henceforth use these chosen versions and omit the hats. Define c_{vu} again by its finite path product and $C_{vu} = (1 - m_v^2)c_{vu}$. These are continuous in time almost surely, and they agree at each fixed time with their former raw versions. Consequently every changed coefficient agrees with its

former version outside a $dt \, d\mathbb{P}$ -null set. Likewise, Fubini and boundedness give equality of the old and new mean integrals on each integer horizon, and hence, after a countable intersection,

$$\int_0^t m_v^0(s, X) \, ds = \int_0^t m_v(s, X) \, ds \quad \text{simultaneously for all } t \geq 0 \quad \text{almost surely.}$$

Thus the innovation paths have not changed.

These versions are also the posteriors for the usual filtrations. For a fixed time t , apply the second part of Lemma 4.2 to the bounded continuous version of $\mathbb{E}[\sigma_v \mid \mathcal{F}_t^0]$. It gives $m_v(t) = \mathbb{E}[\sigma_v \mid \mathcal{F}_t]$. The process $C_{vu} + m_v m_u$ is a bounded continuous version of the raw conditional mean of $\sigma_v \sigma_u$; applying the same argument identifies it with that conditional mean in $\mathcal{F}_t$. This identifies C_{vu} as the augmented conditional covariance. Applied instead to $\tanh h_{v \rightarrow w}$ and the raw component filtration, the argument gives the corresponding assertion for its usual augmentation. These statements include time zero. The one-spin site and component means start at zero. The two-spin posterior starts at its prior value $\mathbb{E}[\sigma_v \sigma_u] = \theta^{\text{dist}(v,u)}$, including value one when $u = v$.

4.5 The field equation

It remains to pass from the posterior mean to its field. This requires the response row $A_v = \sum_u c_{vu}^2$, rather than merely the covariance row: the transformation divides by $1 - m_v^2$. The estimate (3.11), proved before this section, gives

$$\mathbb{E} \int_0^T A_v(s) \, ds < \infty \quad (T < \infty).$$

Its strict case follows from a deterministic geometric sum. At criticality it follows from the intact infinite-side-branch estimate $\mathbb{E} A_v(s) \leq C / \min\{\sqrt{s}, 1\}$ for $s > 0$. This is a different use of attenuation from the finite-boundary Gaussian estimate above. The inputs here are the second-moment information and forward infinite-branch attenuation already proved.

For clarity, define the version used in the drift directly from the chosen observation functions:

$$A_v(s, x) = \begin{cases} \sum_{u \in \mathcal{V}} c_{vu}(s, x)^2, & \text{if this sum is finite,} \\ 0, & \text{if this sum is infinite.} \end{cases}$$

The extended nonnegative sum is a countable sum of Borel progressive functions, so this finite-or-zero version is again a causal Borel progressive observation function. Permuting the summation index and using the same default preserve graph relabeling. The integrated row estimate shows that this version agrees with the extended sum outside a $ds \, d\mathbb{P}$ -null set. The predictable representatives used for Brownian integration agree with the chosen path functions outside one such null set for all u , by countability. Their squared sum therefore agrees with this A_v there. At criticality the unmodified row is infinite at time zero, as noted after (3.11); the displayed convention supplies the value used in the drift.

The Brownian representation and the covariance formula now imply, as an equality of densities up to $ds \, d\mathbb{P}$ -null sets,

$$d\langle m_v \rangle_s = \sum_u C_{vu}(s)^2 \, ds = (1 - m_v(s)^2)^2 A_v(s) \, ds.$$

One may obtain the first equality from finite Brownian sums: their brackets converge in L^1 total variation on every finite horizon, by (4.9), while their martingales converge in $\mathcal{M}_T^2$. The defining bracket identity then passes to the limit.

For $n \geq 2$ let

$$\tau_n = n \wedge \inf\{t \geq 0 : |m_v(t)| \geq 1 - 1/n\}.$$

On the stopped interval the derivatives

$$(\operatorname{atanh})'(x) = \frac{1}{1-x^2}, \quad (\operatorname{atanh})''(x) = \frac{2x}{(1-x^2)^2}$$

are bounded. Itô's formula for the stopped continuous martingale therefore gives

$$h_v(t \wedge \tau_n) = \sum_u \int_0^{t \wedge \tau_n} \frac{C_{vu}(s)}{1 - m_v(s)^2} dW_u(s) + \int_0^{t \wedge \tau_n} m_v(s) A_v(s) ds.$$

The Brownian coefficient is exactly c_{vu} . The equality of this countable Brownian sum with the transformed mean integral follows first for finite sums and then by the isometry; on the stopped interval the derivative is bounded.

The global Brownian sum with coefficients c_{vu} exists in $\mathcal{M}_T^2$. Indeed, for finite $F \subseteq F'$, Doob's inequality gives

$$\mathbb{E} \sup_{0 \leq t \leq T} \left| \sum_{u \in F' \setminus F} \int_0^t c_{vu}(s) dW_u(s) \right|^2 \leq 4 \sum_{u \in F' \setminus F} \mathbb{E} \int_0^T c_{vu}(s)^2 ds.$$

The right side tends to zero along an exhaustion by the integrated row estimate. Also $\mathbb{E} \int_0^T |m_v| A_v ds \leq \mathbb{E} \int_0^T A_v ds < \infty$. Finally $\tau_n \uparrow \infty$ almost surely: the continuous mean stays strictly inside $(-1, 1)$ on every compact interval. The stopped identities hold simultaneously in time by continuity. On any fixed compact interval, almost every path therefore agrees with one of these identities for all sufficiently large n . Removing the stopping yields (4.3).

We now have the full innovation-driven field equation in the observation filtration, with both Brownian and drift terms integrable. The next comparison must prove that the same innovations remain Brownian when two conditional observation histories are put together; that compatibility is the next step of the argument.

5 Comparing observations with the same innovations

The innovations constructed in the preceding section are independent Brownian motions, but that fact alone does not show that they determine the observations. We now turn this question into a comparison of two observation processes. We sample them independently conditional on the same *entire* innovation array, prove that they can still be driven by the same Brownian motions in their joint filtration, and derive an equation for their discrepancy. This is the conditional-copy compatibility method developed by Kurtz [11]; we prove the specific compatibility and transport statements needed here.

Throughout this section, $d \geq 3$, $b = d - 1$, and $0 < \theta = \tanh \beta \leq b^{-1/2}$. We use the continuous causal observation functions selected in Proposition 4.1. With the selected means, the innovation map

$$W_v(t) = X_v(t) - \int_0^t m_v(s) ds$$

is a Borel function of X whose restriction to $[0, t]$ uses only observations up to t . By Section 4 and a countable intersection over vertices, this selected-mean innovation map agrees with the original raw innovation map as a whole-path map almost surely under the observation law. We use this selected version throughout. The field equation (4.3) and the integrated row bound (3.11) have already been proved in the observation filtration. No inverse map from W to X is assumed in what follows. For drift integrals involving A_v , we use the finite-or-zero drift convention fixed in Section 4.

5.1 Conditioning on the complete innovations

Write

$$C_0([0, \infty), \mathbb{R}) = \{x \in C([0, \infty), \mathbb{R}) : x(0) = 0\},$$

$$\mathbf{X} = \mathbf{W} = C_0([0, \infty), \mathbb{R})^{\mathcal{V}}.$$

Give the path space its locally uniform Polish topology and the arrays the countable product topology. The subscript 0 specifies the value at time zero and imposes no condition at infinity. Let $\mathbf{Q}$ be the joint law of (X, W) from the planted observation construction. Its W -marginal is product Wiener measure, denoted by $\mathbf{w}$. These are standard Borel spaces, so there is a regular conditional probability $\mathbf{K}(w, dx)$ for X given $W = w$. Define a probability law on the three path arrays by

$$\widehat{\mathbf{Q}}(dw, dx, dx') = \mathbf{w}(dw) \mathbf{K}(w, dx) \mathbf{K}(w, dx'). \quad (5.1)$$

Write X^1, X^2, W for its coordinates. Each pair (X^j, W) has law $\mathbf{Q}$, while X^1 and X^2 are independent conditional on W . The conditioning in (5.1) uses every coordinate of W at every time, not only the portion already observed.

For $t \geq 0$, let $\mathcal{H}_t^0$ be the raw sigma field generated by $X_v^1(s), X_v^2(s), W_v(s)$ for $v \in \mathcal{V}$ and $0 \leq s \leq t$, and let $(\mathcal{H}_t)$ be its usual augmentation. All posterior fields and coefficients attached to copy j below are obtained by applying the same causal observation functions to X^j .

Proposition 5.1 (Compatible conditional copies). *The law (5.1) is invariant under the simultaneous action of each fixed automorphism of T_d . Every finite subvector of W is standard vector Brownian motion in $(\mathcal{H}_t)$. For each copy j , both equations (4.2) and (4.3) hold in this filtration, with the coefficients of that copy and the common Brownian motions W . Their infinite Brownian sums converge in the maximal L^2 norm on every finite time interval.*

Proof. We first prove that the future of W is not exposed by either observation past. Fix a deterministic $t \geq 0$, write $W_{\leq t}$ for the restriction of W to $[0, t]$, and define its future increment array by

$$Z_v^t(s) = W_v(t + s) - W_v(t), \quad s \geq 0.$$

In the original law $\mathbf{Q}$, Z^t is independent of the raw observation past $\mathcal{F}_t^0$. Indeed, the Brownian property in Proposition 4.1 gives the assertion for every finite set of sites and future times. Rational times generate the Borel sigma field of a continuous path; a monotone-class argument over the countably many sites and rational times gives the assertion for the whole increment array. Causality also makes $W_{\leq t}$ $\mathcal{F}_t^0$ -measurable.

It follows that, for every bounded Borel function F of the observation past,

$$\mathbb{E}_{\mathbf{Q}}[F(X_{\leq t}) \mid W] = \mathbb{E}_{\mathbf{Q}}[F(X_{\leq t}) \mid W_{\leq t}] \quad \text{almost surely.} \quad (5.2)$$

To verify the identity, denote the right side by $J(W_{\leq t})$ and test against $u(W_{\leq t})v(Z^t)$, with bounded Borel u, v . Independence of Z^t from $\mathcal{F}_t^0$ factors $\mathbb{E}v(Z^t)$ out of the expectation containing Fu . It does the same for Ju , and the remaining expectations agree by the definition of J . These products generate $\sigma(W)$: its past together with Z^t reconstructs every path value of W . This proves (5.2).

Conditional independence in (5.1) gives, for bounded past functions F, G ,

$$\mathbb{E}_{\widehat{\mathbf{Q}}}[F(X_{\leq t}^1)G(X_{\leq t}^2) \mid W] = \mathbb{E}_{\mathbf{Q}}[F(X_{\leq t}) \mid W] \mathbb{E}_{\mathbf{Q}}[G(X_{\leq t}) \mid W].$$

Both factors on the right depend only on $W_{\leq t}$ by (5.2). Product Wiener measure makes Z^t independent of $W_{\leq t}$, and hence

$$\begin{aligned} & \mathbb{E}_{\widehat{\mathbf{Q}}}[u(W_{\leq t})F(X_{\leq t}^1)G(X_{\leq t}^2)v(Z^t)] \\ &= \mathbb{E}_{\widehat{\mathbf{Q}}}[u(W_{\leq t})F(X_{\leq t}^1)G(X_{\leq t}^2)] \mathbb{E}_{\mathbf{w}}[v(Z^t)]. \end{aligned}$$

Another monotone-class argument shows that Z^t is independent of the full sigma field $\mathcal{H}_t^0$. Thus every finite subvector of W is Brownian in the raw joined filtration.

Each W_v is a continuous square-integrable martingale in the raw joined filtration and is L^2 -continuous because its marginal law is Wiener measure. Lemma 4.2 therefore makes it a martingale in $(\mathcal{H}_t)$. The quadratic covariations, determined by the Brownian paths, remain $[W_u, W_v]_t = \mathbf{1}_{\{u=v\}}t$. Lévy's martingale characterization applied to each finite subvector proves the claimed Brownian property in $(\mathcal{H}_t)$.

We next check invariance. The selected field functions commute with graph relabeling, so the innovation map does too. Since the observation law is invariant, $\mathbf{Q}$ is invariant under $(X, W) \mapsto (gX, gW)$. For one fixed automorphism g , uniqueness of regular conditional probabilities now gives

$$\mathbf{K}(gw, \cdot) = g_*\mathbf{K}(w, \cdot) \quad \text{for } w\text{-almost every } w. \quad (5.3)$$

To obtain equality as probability measures, first check it for a countable determining algebra generating the Borel sets of $\mathbf{X}$ and intersect the resulting full-measure sets. Applying this identity to both factors in (5.1) proves invariance of the joined law. This argument is made separately for each fixed g .

It remains to justify the stochastic equations in the enlarged filtration. We give the approximation argument because an Itô integral cannot simply be identified across filtrations from a formal equality of pair laws. For either coefficient $k_{vu} = c_{vu}$ or $k_{vu} = C_{vu}$, fix v, u and a finite horizon T . Lemma 4.5, applied to the X -marginal of $\mathbf{Q}$, supplies bounded causal Borel elementary processes

$$G_n(t, X) = \sum_i \gamma_{n,i}(X_{\leq t_{n,i}}) \mathbf{1}_{(t_{n,i}, t_{n,i+1}]}(t), \quad \int_0^T \mathbb{E}_{\mathbf{Q}} |G_n - k_{vu}|^2 dt \longrightarrow 0,$$

where each $\gamma_{n,i}$ is a bounded Borel function of finitely many observations at time $t_{n,i}$, viewed as a function of the indicated past.

The elementary integral against W_u is the Borel path functional

$$I_n(t; X, W) = \sum_i \gamma_{n,i}(X_{\leq t_{n,i}}) (W_u(t \wedge t_{n,i+1}) - W_u(t \wedge t_{n,i})).$$

The Brownian isometry and Doob's maximal inequality show that these integrals converge in $L^2(C([0, T]))$ to the original Itô integral. A deterministic subsequence converges uniformly almost surely; its limit on the Borel set of uniformly Cauchy inputs, with the zero path as default, is a Borel version of that integral as a function of (X, W) . The same subsequence has that pathwise limit in each joined pair, since (X^j, W) has law $\mathbf{Q}$. On the other hand the same isometry and maximal inequality in $(\mathcal{H}_t)$ show that $I_n(\cdot; X^j, W)$ converges to the joined Itô integral of $k_{vu}(\cdot, X^j)$. The two limits agree. This proves transport of each individual integral, including its continuous path version.

For either coefficient family the integrated squared row is finite: this is (3.11) for c , and follows for C also from $C_{vu} = (1 - m_v^2)c_{vu}$. If $F \subset F'$ are finite vertex sets, the joined integral tails satisfy

$$\mathbb{E}_{\widehat{\mathbf{Q}}} \sup_{t \leq T} \left| \sum_{u \in F' \setminus F} \int_0^t k_{vu}(s, X^j) dW_u(s) \right|^2 \leq 4 \sum_{u \in F' \setminus F} \mathbb{E}_{\mathbf{Q}} \int_0^T k_{vu}(s, X)^2 ds.$$

The right side tends to zero along an exhaustion. The drift at any fixed time is a Borel observation functional, obtained by time integration; its absolute integral is finite almost surely because $|m_v| \leq 1$ and $\mathbb{E} \int_0^T A_v dt < \infty$. Assigning zero to exceptional nonfinite integral values supplies a Borel scalar version if needed. Transport the original equations with finite partial Brownian sums at rational

times and let the partial sums increase: their residuals converge in probability under $\mathbf{Q}$, hence under each identical joined pair law, while their joined integral limits have just been identified. The equations therefore hold at all rational times in the joining. Continuity of the fields and Brownian sums, and absolute continuity of the drift integrals, extend them to all times. Taking countable intersections over vertices and integer horizons finishes the transport assertion. $\square$

The purpose of this construction can already be stated precisely. For each vertex v and nonnegative rational q , $\mathbb{E}_{\mathbf{Q}} X_v(q)^2 = q + q^2 < \infty$. Conditional independence gives

$$\mathbb{E}_{\hat{\mathbf{Q}}}[(X_v^1(q) - X_v^2(q))^2 \mid W] = 2 \operatorname{Var}_{\mathbf{Q}}(X_v(q) \mid W) \quad \text{almost surely.} \quad (5.4)$$

Thus equality of the two observation paths would make every rational coordinate have zero conditional variance. These countably many coordinates determine a continuous path configuration. The final section will turn that implication into a Borel recovery map after the comparison has proved equality; no equality has been assumed here.

5.2 The field discrepancy and its variance

From now on unprimed quantities refer to the first copy, primed quantities to the second, and Δ means their difference. All expectations in the comparison are under $\hat{\mathbf{Q}}$. Every one-copy estimate proved earlier remains valid because the marginals have not changed.

Fix $T > 0$. In each pair the defining innovation identity holds as an identity of continuous paths. Subtracting the two identities gives

$$\Delta X_v(t) = \int_0^t \Delta m_v(s) \, ds, \quad |\Delta X_v(t)| \leq 2t.$$

Together with $|h_v - X_v|, |h'_v - X'_v| \leq d\beta$, this gives

$$|\Delta h_v(t)| \leq H_T := 2T + 2d\beta \quad (v \in \mathcal{V}, 0 \leq t \leq T) \quad (5.5)$$

on one full-probability event. The field bounds and identities hold simultaneously because the chosen fields are continuous and the vertices are countable.

Choose a vertex o only for taking expectations, and put

$$V(t) = \mathbb{E}[(\Delta h_o(t))^2].$$

Invariance of the joined law makes the value independent of the choice of o . It is nonnegative, bounded by H_T^2 on $[0, T]$, and $V(0) = 0$. We need the full variance equation, including its drift.

Lemma 5.2 (Variance of the discrepancy). *The function V is absolutely continuous on every finite interval. For almost every t ,*

$$V'(t) = \sum_{u \in \mathcal{V}} \mathbb{E}[(\Delta c_{ou}(t))^2] + 2\mathbb{E}[\Delta h_o(t)(m_o(t)A_o(t) - m'_o(t)A'_o(t))]. \quad (5.6)$$

Proof. Subtract the two field equations in the common filtration. The martingale part is $M_o = \sum_u \int \Delta c_{ou} \, dW_u$, and, outside the $dt \, d\hat{\mathbf{Q}}$ -null set where a row diverges,

$$\sum_u (\Delta c_{ou})^2 \leq 2(A_o + A'_o).$$

The integrated expectation of the right side is finite. Thus the sum defining M_o converges in maximal L^2 , and its bracket is

$$\langle M_o \rangle_t = \int_0^t \sum_u (\Delta c_{ou}(s))^2 ds.$$

This follows first for finite sums from the zero cross-covariations of distinct Brownian coordinates, then for the limit by the L^1 convergence of the integrated squared row tails. Itô's formula for $(\Delta h_o)^2$ gives

$$\begin{aligned} (\Delta h_o(t))^2 &= 2 \int_0^t \Delta h_o(s) dM_o(s) + \int_0^t \sum_u (\Delta c_{ou}(s))^2 ds \\ &\quad + 2 \int_0^t \Delta h_o(s) (m_o(s) A_o(s) - m'_o(s) A'_o(s)) ds. \end{aligned}$$

The first integral is a square-integrable martingale: by (5.5), its expected bracket is at most $8H_T^2 \mathbb{E} \int_0^T (A_o + A'_o) ds$. The absolute values of the other two integrands have expectations bounded respectively by $2\mathbb{E}(A_o + A'_o)$ and $2H_T \mathbb{E}(A_o + A'_o)$, which are integrable in time. Taking expectations therefore gives an integral of an $L^1([0, T])$ function. This proves absolute continuity and its almost-everywhere derivative (5.6). $\square$

The coefficients in this equation are products along directed paths. We first control the difference of one factor. For a directed edge $e = (i, j)$ define

$$D_e(t) = \log a(h_{j \rightarrow i}(t)) - \log a(h'_{j \rightarrow i}(t)), \quad S(t) = \mathbb{E}[D_e(t)^2]. \quad (5.7)$$

Transitivity on directed edges makes S independent of e . The two cavity identities at the ends of ij give

$$|\Delta h_{j \rightarrow i}| \leq |\Delta h_j| + \theta |\Delta h_{i \rightarrow j}|, \quad |\Delta h_{i \rightarrow j}| \leq |\Delta h_i| + \theta |\Delta h_{j \rightarrow i}|.$$

Substituting the second inequality into the first yields

$$|\Delta h_{j \rightarrow i}| \leq \frac{|\Delta h_j| + \theta |\Delta h_i|}{1 - \theta^2}. \quad (5.8)$$

In particular, on $[0, T]$ it is bounded by $H_T/(1 - \theta)$; Minkowski's inequality and invariance also give $\|\Delta h_{j \rightarrow i}\|_2 \leq \sqrt{V}/(1 - \theta)$.

Direct differentiation gives

$$(\log a)'(z) = -\frac{2(1 - \theta^2) \tanh z}{1 - \theta^2 \tanh^2 z}, \quad |(\log a)'(z)| \leq 2|\tanh z| \leq 2. \quad (5.9)$$

Here the first inequality uses $1 - \theta^2 \tanh^2 z \geq 1 - \theta^2$. We conclude that

$$|D_e(t)| \leq \frac{2H_T}{1 - \theta}, \quad S(t) \leq \frac{4}{(1 - \theta)^2} V(t) \quad (0 \leq t \leq T). \quad (5.10)$$

Also $(\log a)(0) = 0$, so $|\log a(h_{j \rightarrow i}(t))| \leq 2|h_{j \rightarrow i}(t)| \leq 2(|X_j(t)| + b\beta)$. Since $\mathbb{E}X_j(t)^2 = t + t^2$, each one-copy log factor is square integrable. Equality of the marginal laws then gives $\mathbb{E}D_e(t) = 0$. These facts will permit the later spatial cancellation estimate to be applied to this particular edge observable.

5.3 Reducing the comparison to path estimates

For a directed simple path $v_0, \dots, v_r$, $r \geq 1$, let $e_i = (v_{i-1}, v_i)$. At a common time t , set

$$\begin{aligned} P_r &= \prod_{i=1}^r a(h_{v_i \rightarrow v_{i-1}}), & Z_r &= \sum_{i=1}^r D_{e_i}, \\ Q_r &= \mathbb{E}[(P_r^2 + (P'_r)^2)Z_r^2], & L_r &= \mathbb{E}[(\Delta h_{v_0})^2 P_r^2]. \end{aligned} \quad (5.11)$$

The orientation matters: every factor uses the component at the next vertex excluding the preceding one. Invariance makes these expectations depend only on r and t , so we suppress the particular path. Both P_r and P'_r lie in $(0, 1]$, and $\log P_r - \log P'_r = Z_r$. The mean-value theorem, first for e^x and then for e^{2x} , gives

$$\mathbb{E}[(P_r - P'_r)^2] \leq 2Q_r, \quad |P_r^2 - (P'_r)^2| \leq 2(P_r^2 + (P'_r)^2)|Z_r|. \quad (5.12)$$

For example, the derivative of e^x on the segment joining $\log P_r$ and $\log P'_r$ is at most $P_r + P'_r$; squaring and using $(P_r + P'_r)^2 \leq 2(P_r^2 + (P'_r)^2)$ proves the first bound. The derivative of e^{2x} on the same segment is at most $2(P_r^2 + (P'_r)^2)$, proving the second.

There are $n_r = db^{r-1}$ vertices at distance $r \geq 1$ from o . The diagonal coefficients are $c_{oo} = c'_{oo} = 1$, and the coefficient along any positive-length path is $\theta^r P_r$. Tonelli's theorem and (5.12) therefore bound the noise term in (5.6) by

$$\sum_u \mathbb{E}[(\Delta c_{ou})^2] \leq 2 \sum_{r \geq 1} n_r \theta^{2r} Q_r. \quad (5.13)$$

The drift contributes a second, different path sum:

$$|\mathbb{E}[\Delta h_o(m_o A_o - m'_o A'_o)]| \leq V + \sum_{r \geq 1} n_r \theta^{2r} (L_r + 2\sqrt{2V}\sqrt{Q_r}). \quad (5.14)$$

To prove this, write $m_o A_o - m'_o A'_o = (m_o - m'_o)A_o + m'_o(A_o - A'_o)$. Since $\tanh$ is 1-Lipschitz, the first part is bounded by

$$\mathbb{E}[(\Delta h_o)^2 A_o] = V + \sum_{r \geq 1} n_r \theta^{2r} L_r.$$

For the second, use $|m'_o| \leq 1$. The diagonal terms of $A_o - A'_o$ cancel. For each remaining path, put $w_r = P_r^2 + (P'_r)^2 \leq 2$. The second inequality in (5.12) and weighted Cauchy–Schwarz give

$$\begin{aligned} \mathbb{E}[|\Delta h_o| |P_r^2 - (P'_r)^2|] &\leq 2\sqrt{\mathbb{E}[(\Delta h_o)^2 w_r]} \sqrt{\mathbb{E}[w_r Z_r^2]} \\ &\leq 2\sqrt{2V}\sqrt{Q_r}. \end{aligned}$$

Summing proves (5.14). These inequalities may first be read for finite spatial sums and then with nonnegative extended right sides; the estimates at their subsequent uses make those right sides finite. In particular no independence between a path product and its discrepancy is used. The square-root sum in the drift must be controlled along with the noise sum.

Proposition 5.3 (Uniqueness below the threshold). *If $b\theta^2 < 1$, then $V(t) = 0$ for every $t \geq 0$.*

Proof. Since the products in (5.11) are at most one, $L_r \leq V$. Cauchy–Schwarz in the sum Z_r gives

$$Q_r \leq 2\mathbb{E}Z_r^2 \leq 2r \sum_{i=1}^r \mathbb{E}D_{e_i}^2 = 2r^2 S.$$

Use this with (5.10) in the noise and drift bounds. The contributions are bounded by V times sums of r^2 , 1, and r , respectively, against

$$n_r \theta^{2r} = \frac{d}{b} (b\theta^2)^r.$$

All three series converge when $b\theta^2 < 1$. Equation (5.6) therefore gives $V' \leq CV$ almost everywhere on $[0, T]$, for a finite constant depending on the fixed parameters. Absolute continuity and $V(0) = 0$ imply $V = 0$ there by Gronwall's inequality. The horizon T was arbitrary. $\square$

At equality, $n_r \theta^{2r} = d/b$, so the geometric decay in this proof disappears. The next two sections will replace the elementary $r^2 S$ loss by spatial cancellation of order rS and attenuation on the length scale $\omega(t)^{-1}$. Their leading noise estimate has the form $Q_r \lesssim rS(t)e^{-c\omega(t)r}$. Near time zero, where $\omega(t) = \sqrt{t}$, its sum has size

$$S(t) \sum_{r \geq 1} r e^{-c\sqrt{t}r} = S(t) \frac{e^{-c\sqrt{t}}}{(1 - e^{-c\sqrt{t}})^2} \asymp \frac{S(t)}{t}.$$

Thus even those two estimates, combined only with $S \leq CV$, would leave a nonintegrable coefficient at time zero. The final comparison will improve the relation between S and V using small component fields, while also retaining the square-root drift sum. This is the remaining obstruction at the critical parameter.

6 Cancellation along a path

The comparison of the two observation copies led to sums of the edge differences $D_e(t)$. Cauchy–Schwarz bounds the second moment of a sum over r edges by $r^2 S(t)$, where $S(t) = \mathbb{E} D_e(t)^2$. We now improve the factor r^2 to r by proving that the correlations of these differences are summable along a path. The proof begins with local correlations under one observation law and passes to the joining through the closed space generated by differences of marginal functions. Although only equality remains in the comparison, this path estimate holds throughout $0 < \theta \leq b^{-1/2}$.

Here and below ν is the observation marginal of the original joint law $\mathbf{Q}$, namely the law of the entire array

$$X = (X_v)_{v \in \mathcal{V}} \quad \text{in} \quad \mathbf{X} = C_0([0, \infty), \mathbb{R})^{\mathcal{V}}.$$

Here C_0 again means continuous paths starting at zero, with the topology of uniform convergence on compact time intervals. The vertex and nonbacktracking operator geometry used in this argument is related to the correlation arguments of Backhausz, Szegedy, and Virág [1, Theorem 2.2 in arXiv:1305.6784v1] and Backhausz, Gerencsér, Harangi, and Vizer [2, Theorems 4.1–4.2 in arXiv:1603.08423v2]. The proof below supplies the precise transfer to differences of marginal observation functions.

Proposition 6.1 (Path variance). *Let $d \geq 3$, $b = d - 1$, and $0 < \theta \leq b^{-1/2}$. Let π be a probability law on $\mathbf{X}^2$ with both marginals equal to ν , invariant under the simultaneous action of $\text{Aut}(T_d)$. Suppose that $g = (g_e)$ assigns a measurable real function of $\mathbf{X}$ to every directed edge, with $\mathbb{E}_\nu g_e^2 < \infty$, and that for each fixed $\gamma \in \text{Aut}(T_d)$ and directed edge e ,*

$$g_{\gamma e}(\gamma x) = g_e(x) \quad \text{for } \nu\text{-almost every } x.$$

Here $\gamma(i, j) = (\gamma i, \gamma j)$. For the two coordinate arrays X, X' under π , set

$$D_e = g_e(X) - g_e(X'), \quad S = \mathbb{E}_\pi D_e^2.$$

The value of S is independent of the directed edge. There is a constant $C_d < \infty$, depending only on d , such that for every integer $r \geq 1$ and every simple path $v_0, \dots, v_r$,

$$\begin{aligned} \mathbb{E}_\pi \left[\left(\sum_{i=1}^r D_{(v_{i-1}, v_i)} \right)^2 \right] &\leq C_d r S, \\ \mathbb{E}_\pi \left[\left(\sum_{i=1}^r D_{(v_i, v_{i-1})} \right)^2 \right] &\leq C_d r S. \end{aligned} \tag{6.1}$$

In particular, take π to be the (X, X') -marginal of the joining in Proposition 5.1. Then this holds for the differences $D_e(t)$ in (5.7) at each fixed finite time, with $S = S(t)$.

We prove the proposition in three steps. First, local observation functions have exponentially small correlations across a long connecting path. This bounds the self-adjoint neighbor average on centered vertex functions. Second, the bound transfers to a closed space of differences inside the joining. Finally, head and tail lifts of that vertex space control the nonbacktracking edge average. Its powers are summable, which gives the linear variance in (6.1). All equivariance identities in this section are understood separately for each fixed automorphism, as in the proposition.

6.1 The vertex bound for one observation law

We shall use the same Hilbert-space notation for several invariant configuration laws. For such a law η , a *vertex assignment* is a family $f = (f_v)_{v \in \mathcal{V}}$ of measurable real functions satisfying

$$f_{\gamma v}(\gamma x) = f_v(x) \quad \eta\text{-almost surely, for each fixed } \gamma, v.$$

Let $\mathcal{H}_v(\eta)$ be the space of square-integrable vertex assignments, modulo almost sure equality at each vertex, with

$$\langle f, k \rangle_{v, \eta} = \mathbb{E}_\eta[f_o k_o].$$

Transitivity and invariance make this independent of the vertex o . The analogous space $\mathcal{H}_e(\eta)$ consists of assignments to directed edges, with inner product at any one directed edge. These spaces are Hilbert spaces: a Cauchy sequence has an L^2 limit at each index, and the equivariance identity for any fixed automorphism passes to the limits because that automorphism preserves η . There are countably many indices, so equality in the indicated norm is equality of assignments. We use real spaces; the usual spectral theorem applies after complexification if needed.

On $\mathcal{H}_v(\eta)$ define the neighbor average

$$(\mathbf{A}_\eta f)_v = \frac{1}{d} \sum_{u \sim v} f_u.$$

Jensen's inequality and invariance give $\|\mathbf{A}_\eta\| \leq 1$. For adjacent o, u , an automorphism interchanging them gives $\mathbb{E}_\eta[f_o k_u] = \mathbb{E}_\eta[f_u k_o]$. Averaging this identity over the neighbors of o proves that $\mathbf{A}_\eta$ is self-adjoint. It preserves the closed subspace of centered assignments $\mathbb{E}_\eta f_o = 0$. This subspace is reducing: if y is orthogonal to it, then $\langle \mathbf{A}_\eta y, f \rangle = \langle y, \mathbf{A}_\eta f \rangle = 0$ for every centered f . Thus the restriction to that subspace is again self-adjoint.

Lemma 6.2 (Marginal vertex spectrum). *For the full observation law ν , the spectrum of $\mathbf{A}_\nu$ on centered vertex assignments is contained in $[-\rho, \rho]$, where*

$$\rho = \frac{2\sqrt{b}}{d} < 1. \tag{6.2}$$

Proof. We first prove a correlation bound for an assignment f that is centered and measurable, at vertex v , from the entire observation paths at the vertices in a fixed ball $B_R(v)$. Square integrability is enough; write $\|f\|^2 = \mathbb{E}_\nu f_o^2$. Write $\mathbb{E}_{\text{pl}}$ for expectation in the planted spin-and-Brownian construction, whose observation marginal is ν .

Suppose that $\text{dist}(u, v) = k > 2R$, and let a and c be the respective endpoints of the path joining $B_R(u)$ to $B_R(v)$. Then $\text{dist}(a, c) = k - 2R$. In the planted broadcast construction, conditional on (σ_a, σ_c) , the observation arrays in these two balls are independent. The conditional law in the first ball depends only on σ_a , and the law in the second only on σ_c . To see this, root the broadcast at a . Once the endpoint spins are fixed, the edge multipliers inside the two balls are taken from disjoint branches off the connecting path. They are independent of that path and of one another, and the Brownian paths at the two disjoint sets of vertices are independent as well. The assertion for products of square-integrable functions follows from its bounded version by truncation.

The endpoint spins are uniform signs. Since f_u, f_v are centered, their conditional expectations have the form

$$\mathbb{E}_{\text{pl}}[f_u \mid \sigma_a] = \alpha_u \sigma_a, \quad \mathbb{E}_{\text{pl}}[f_v \mid \sigma_c] = \alpha_v \sigma_c.$$

L^2 -contraction of conditional expectation gives $|\alpha_u|, |\alpha_v| \leq \|f\|$. The broadcast correlation along the connecting path is $\mathbb{E}_{\text{pl}}[\sigma_a \sigma_c] = \theta^{k-2R}$. It follows that

$$|\mathbb{E}_\nu[f_u f_v]| = |\alpha_u \alpha_v| \theta^{k-2R} \leq \|f\|^2 b^R b^{-k/2}.$$

For $k \leq 2R$ the same bound follows from Cauchy-Schwarz, since $b^R b^{-k/2} \geq 1$. We have therefore proved

$$|\mathbb{E}_\nu[f_u f_v]| \leq C_f b^{-\text{dist}(u,v)/2} \quad \text{for all } u, v, \quad C_f = b^R \|f\|^2. \quad (6.3)$$

This calculation uses the planted spins to describe the law, while the assignment itself remains a function of the observations alone.

We now turn spatial decay into an operator bound. Let $p_n(o, v)$ be the n -step transition probabilities of simple random walk on the tree and put $q(v) = b^{-\text{dist}(o,v)/2}$. For the deterministic neighbor average P on functions of vertices,

$$(Pq)(v) = \rho q(v) \quad (v \neq o), \quad (Pq)(o) = b^{-1/2} \leq \rho q(o).$$

The first identity uses one neighbor closer to o and b farther away. Positivity of P gives $P^n q \leq \rho^n q$ by induction, and hence

$$\sum_v p_n(o, v) q(v) \leq \rho^n.$$

Only finite-step random-walk averages are used here; q need not belong to $\ell^2(\mathcal{V})$. Self-adjointness of A_ν , its finite-walk expansion, and (6.3) now give, for every $n \geq 0$,

$$\|A_\nu^n f\|^2 = \langle f, A_\nu^{2n} f \rangle = \sum_v p_{2n}(o, v) \mathbb{E}_\nu[f_o f_v] \leq C_f \rho^{2n}. \quad (6.4)$$

Let μ_f be the spectral measure of f for the self-adjoint restriction of A_ν to centered assignments. For $\varepsilon > 0$, (6.4) implies

$$\mu_f\{|\lambda| \geq \rho + \varepsilon\} \leq C_f \left(\frac{\rho}{\rho + \varepsilon} \right)^{2n}.$$

Letting $n \rightarrow \infty$ proves that μ_f is supported on $[-\rho, \rho]$.

It remains to pass from local assignments to all centered ones. For $f \in \mathcal{H}_\nu(\nu)$ centered, let

$$f_v^{(R)} = \mathbb{E}_\nu[f_v \mid \sigma\{X_z(s) : z \in B_R(v), s \geq 0\}].$$

These are centered square-integrable local assignments. To verify their equivariance, pull the conditional expectation at γv back by a fixed automorphism γ . Invariance of ν pulls its conditioning sigma-field back to that for $B_R(v)$, and equivariance pulls its integrand back to f_v . Uniqueness of conditional expectation gives the required identity almost surely for that γ, v . As $R \rightarrow \infty$, the conditioning sigma-fields generate the complete observation array, so $f_o^{(R)} \rightarrow f_o$ in L^2 . This proves density of the local assignments just used. Equally, truncating and recentering f first gives a bounded local dense family. The range of the corresponding spectral projection $\mathbf{1}_{[-\rho, \rho]}(\mathbf{A}_\nu)$ on the centered space is closed and contains every such local assignment. It is consequently the entire centered space, proving the lemma. Finally, $\rho < 1$ follows from $d = b + 1$ and $(b + 1)^2 - 4b = (b - 1)^2 > 0$. $\square$

6.2 Passing the bound to marginal differences

Fix the joining π in Proposition 6.1. For a marginal vertex or edge assignment define

$$(\delta f)(X, X') = f(X) - f(X').$$

This is well defined on the marginal L^2 equivalence classes: each coordinate under π has law ν . The same observation and the triangle inequality give $\|\delta f\|_\pi \leq 2\|f\|_\nu$. In the joined assignment spaces put

$$\mathcal{K} = \overline{\delta \mathcal{H}_v(\nu)} \subseteq \mathcal{H}_v(\pi), \quad \mathcal{D} = \overline{\delta \mathcal{H}_e(\nu)} \subseteq \mathcal{H}_e(\pi),$$

where the closures use the respective $L^2(\pi)$ norms. These are closed linear spaces.

The vertex neighbor averages for ν and π commute with δ , since they use the same finite sums over neighboring vertex indices. Thus $\mathbf{A}_\pi \mathcal{K} \subseteq \mathcal{K}$. The joined $\mathbf{A}_\pi$ is self-adjoint, so the same orthogonality argument as above shows that $\mathcal{K}$ is reducing and its restriction $\mathbf{A}_\mathcal{K} := \mathbf{A}_\pi|_{\mathcal{K}}$ is self-adjoint.

Subtracting the constant $\mathbb{E}_\nu f_o$ from a marginal vertex assignment does not change its difference. Lemma 6.2 therefore gives

$$\|\mathbf{A}_\pi^n \delta f\|_\pi = \|\delta(\mathbf{A}_\nu^n(f - \mathbb{E}_\nu f_o))\|_\pi \leq 2\rho^n \|f - \mathbb{E}_\nu f_o\|_\nu \quad (n \geq 0). \quad (6.5)$$

Applying the spectral-measure argument from (6.4) to the joined self-adjoint operator $\mathbf{A}_\pi$ shows that the spectral measure of δf is supported on $[-\rho, \rho]$. The constant on the right of (6.5) may depend on the chosen preimage f ; only its exponential rate matters as $n \rightarrow \infty$. In particular, no lower comparison between $\|\delta f\|_\pi$ and $\|f - \mathbb{E}_\nu f_o\|_\nu$ is used. The closed range of the joined spectral projection $\mathbf{1}_{[-\rho, \rho]}(\mathbf{A}_\pi)$ contains the closure of all these differences. Hence

$$\|\mathbf{A}_\mathcal{K}\| \leq \rho. \quad (6.6)$$

This is the vertex estimate needed below. Its domain is $\mathcal{K}$, the closed space generated by marginal differences.

6.3 From vertices to nonbacktracking edge averages

A vertex assignment can be read at either endpoint of a directed edge. We compare these two lifts and then isolate the part of an edge assignment orthogonal to both. On the joined assignment spaces define the tail and head lifts $\mathbf{F}, \mathbf{H} : \mathcal{H}_v(\pi) \rightarrow \mathcal{H}_e(\pi)$, the reversal $\mathbf{J}$, and the forward nonbacktracking average $\mathbf{N}$ by

$$\begin{aligned} (\mathbf{F}f)_{(i,j)} &= f_i, & (\mathbf{H}f)_{(i,j)} &= f_j, & (\mathbf{J}g)_{(i,j)} &= g_{(j,i)}, \\ (\mathbf{N}g)_{(i,j)} &= \frac{1}{b} \sum_{\substack{k \sim j \\ k \neq i}} g_{(j,k)}. \end{aligned}$$

Invariance makes the two lifts isometries and makes J a self-adjoint isometry with $J^2 = I$. Jensen's inequality makes N a contraction. The adjoints of the lifts are

$$(F^*g)_v = \frac{1}{d} \sum_{u \sim v} g(v,u), \quad (H^*g)_v = \frac{1}{d} \sum_{u \sim v} g(u,v). \quad (6.7)$$

Indeed, in the edge inner product we may average the fixed outgoing edge over all neighbors of its tail, because the tail stabilizer is transitive on those neighbors. This proves the first formula. Interchanging the endpoints first proves the second. In particular,

$$F^*H = H^*F = A_\pi, \quad JF = H, \quad JH = F.$$

Direct substitution in the definitions gives

$$N = \frac{d}{b} HF^* - \frac{1}{b} J, \quad N^* = \frac{d}{b} FH^* - \frac{1}{b} J, \quad (6.8)$$

$$NF = H, \quad NH = \frac{d}{b} HA_\pi - \frac{1}{b} F. \quad (6.9)$$

Thus $(N^*g)_{(i,j)} = b^{-1} \sum_{k \sim i, k \neq j} g(k,i)$ and $N^* = JNJ$.

The same formulas act on the marginal assignment spaces. They use finite sums over neighboring indices, selections of endpoint values, and edge reversal, so they commute with taking marginal differences. Apply this first on the ranges of δ , and then use boundedness to pass to their closures. We obtain

$$FK, HK \subseteq \mathcal{D}, \quad F^*\mathcal{D}, H^*\mathcal{D} \subseteq \mathcal{K}, \quad J\mathcal{D} = \mathcal{D}.$$

Equation (6.8) now shows that both N and N^* preserve $\mathcal{D}$. Thus $\mathcal{D}$ is a reducing subspace for the nonbacktracking operator.

Define

$$R : \mathcal{K} \oplus \mathcal{K} \longrightarrow \mathcal{D}, \quad R(f, g) = Ff + Hg.$$

The adjoint identities and (6.6) give

$$\begin{aligned} \|R(f, g)\|^2 &= \|f\|^2 + \|g\|^2 + 2\langle f, A_{\mathcal{K}}g \rangle, \\ (1 - \rho)(\|f\|^2 + \|g\|^2) &\leq \|R(f, g)\|^2 \leq (1 + \rho)(\|f\|^2 + \|g\|^2). \end{aligned} \quad (6.10)$$

Here $2|\langle f, A_{\mathcal{K}}g \rangle| \leq \rho(\|f\|^2 + \|g\|^2)$. Because $\rho < 1$, R is injective with closed range

$$\mathcal{L} = FK + HK.$$

In fact, the lower bound makes the preimages of a convergent sequence in the range Cauchy. It also gives $\|R^{-1}\| \leq (1 - \rho)^{-1/2}$ on $\mathcal{L}$, while the upper bound gives $\|R\| \leq (1 + \rho)^{1/2}$.

For $y \in \mathcal{D} \ominus \mathcal{L}$, the vectors F^*y, H^*y lie in $\mathcal{K}$ and are orthogonal to every vector in $\mathcal{K}$; for example $\langle F^*y, f \rangle = \langle y, Ff \rangle = 0$. Both therefore vanish. Reversal preserves $\mathcal{D}$ and interchanges the two lifts in $\mathcal{L}$, so it preserves $\mathcal{D} \ominus \mathcal{L}$ as well. On this complement, (6.8) reduces to

$$Ny = -b^{-1}Jy, \quad \|N^n y\| = b^{-n}\|y\| \quad (n \geq 0). \quad (6.11)$$

The space $\mathcal{L}$ is also invariant under N by (6.9). We have split $\mathcal{D}$ into two orthogonal invariant spaces. The complement already decays geometrically; it remains to bound powers on the head and tail range.

On $\mathcal{K} \oplus \mathcal{K}$, (6.9) reads

$$\mathbf{R}^{-1}\mathbf{N}\mathbf{R} = \mathbf{T} = \begin{pmatrix} 0 & -b^{-1}I \\ I & (d/b)\mathbf{A}_{\mathcal{K}} \end{pmatrix}.$$

This matrix need not be normal. We bound its powers by its second-order recurrence. Put

$$\mathbf{Z} = \sqrt{b}\mathbf{T}, \quad \mathbf{B} = \frac{d}{2\sqrt{b}} \begin{pmatrix} \mathbf{A}_{\mathcal{K}} & 0 \\ 0 & \mathbf{A}_{\mathcal{K}} \end{pmatrix}.$$

The operator $\mathbf{B}$ is self-adjoint and $\|\mathbf{B}\| \leq 1$. Matrix multiplication gives

$$\mathbf{Z}\mathbf{B} = \mathbf{B}\mathbf{Z}, \quad \mathbf{Z}^2 - 2\mathbf{B}\mathbf{Z} + I = 0. \quad (6.12)$$

Let $U_{-1} = 0$, $U_0 = 1$, and $U_{m+1}(x) = 2xU_m(x) - U_{m-1}(x)$ be the Chebyshev polynomials of the second kind. Induction using (6.12) gives

$$\mathbf{Z}^n = U_{n-1}(\mathbf{B})\mathbf{Z} - U_{n-2}(\mathbf{B}) \quad (n \geq 1).$$

For $m \geq 0$, the trigonometric identity

$$U_m(\cos s) = \sum_{j=0}^m e^{i(m-2j)s}$$

gives $\sup_{[-1,1]} |U_m| \leq m+1$, including the endpoints. The spectral theorem applied to the self-adjoint contraction $\mathbf{B}$ therefore yields

$$\|\mathbf{Z}^n\| \leq n\|\mathbf{Z}\| + (n-1) \quad (n \geq 1).$$

For completeness, the off-diagonal part of $\mathbf{Z}$ has norm $\sqrt{b}$, while its only nonzero diagonal block is $(d/\sqrt{b})\mathbf{A}_{\mathcal{K}}$, of norm at most 2. Thus $\|\mathbf{Z}\| \leq \sqrt{b} + 2$. At the allowed scalar endpoints ± 1 for $\mathbf{B}$, the characteristic polynomial in (6.12) has a double root. The linear factor in n accounts for this endpoint behavior.

Using $\mathbf{T}^n = b^{-n/2}\mathbf{Z}^n$ and the two bounds for $\mathbf{R}$ and $\mathbf{R}^{-1}$, we conclude that $\|\mathbf{N}^n|_{\mathcal{L}}\| \leq C_d(n+1)b^{-n/2}$. Together with (6.11) and the orthogonal invariant decomposition, this gives

$$\|\mathbf{N}^n|_{\mathcal{D}}\| \leq C_d(n+1)b^{-n/2} \quad (n \geq 0). \quad (6.13)$$

The same estimate holds for $(\mathbf{N}^*)^n$ on $\mathcal{D}$, by reversal. The spectral theorem was used only for the self-adjoint vertex averages and $\mathbf{B}$; the edge estimate follows from the displayed recurrence.

6.4 Summing the edge correlations

Let $D = \delta g \in \mathcal{D}$ be the difference in the proposition. Its mean at every directed edge is zero, since the two marginals are equal, and $\|D\|^2 = S$. Fix a directed simple path and write $e_i = (v_{i-1}, v_i)$. If $i < j$, invariance and transitivity on directed simple paths give

$$\mathbb{E}_{\pi}[D_{e_i}D_{e_j}] = \langle D, \mathbf{N}^{j-i}D \rangle_{e,\pi}. \quad (6.14)$$

Indeed, $\mathbf{N}^{j-i}$ averages over the b^{j-i} nonbacktracking continuations of a fixed directed edge. On a tree each such continuation is simple. An automorphism carries it to the corresponding fixed subpath, so every term has the expectation on the left. This is a finite average of identities for fixed automorphisms.

By (6.13), the absolute value of the lag- n correlation is at most $C_d(n+1)b^{-n/2}S$. Expanding the square of the sum and grouping equal lags therefore gives

$$\begin{aligned}\mathbb{E}_\pi \left[\left(\sum_{i=1}^r D_{e_i} \right)^2 \right] &= rS + 2 \sum_{n=1}^{r-1} (r-n) \langle D, \mathbf{N}^n D \rangle \\ &\leq rS \left(1 + 2C_d \sum_{n=1}^{\infty} (n+1)b^{-n/2} \right) \leq C'_d rS.\end{aligned}$$

The series converges because $b \geq 2$. If all edges are reversed, order them as $(v_r, v_{r-1}), \dots, (v_1, v_0)$. They are the forward edges of the reversed simple path, and their sum is unchanged by this reordering. The same argument proves the second inequality of (6.1).

Finally, take π to be the (X, X') -marginal of the joining in Proposition 5.1, forgetting its common innovation coordinate. At a fixed finite time t , use

$$g_{(i,j)}(X) = \log a(h_{j \rightarrow i}(t)).$$

The component versions constructed earlier make this an equivariant marginal assignment. The cavity field bound and $(\log a)(0) = 0$, $|(\log a)'| \leq 2$, imply

$$|g_{(i,j)}(X)| \leq 2|h_{j \rightarrow i}(t)| \leq 2(|X_j(t)| + b\beta).$$

Since $\mathbb{E}_\nu X_j(t)^2 = t + t^2$, it is square integrable. Its difference is precisely $D_{(i,j)}(t)$. Applying the general estimate completes the proof of Proposition 6.1. The bound is uniform in the time and in the joining because its constant depends only on d .

7 Attenuation with dependent path weights

The path variance estimate controls sums of edge discrepancies, but the critical comparison also needs decay of the products multiplying those sums. Those products and differences depend on the same observations. We therefore average over paths before using the variance estimate. Keeping the first half of a path fixed gives an outward continuation; keeping the second half fixed requires enumerating the first half backward. This is why we need concentration in both directions.

Throughout this section, $b = d - 1 \geq 2$, $\theta = b^{-1/2}$, and $\omega(t) = \min\{\sqrt{t}, 1\}$. We first fix $t > 0$ and work with the spins and observations in the planted realization of one marginal. All fields below are evaluated at this time, which we usually suppress. In particular, $h_{v \rightarrow w}$ is the previously chosen measurable function of the observation history in $T_{v \rightarrow w}$ alone. We use the shifted estimate from one branch (3.7): for an intact infinite component attached to a spin σ_w , every real z , and either value s of that spin,

$$\mathbb{E} \left[a(z + \psi(h_{v \rightarrow w}))^2 \mid \sigma_w = s \right] \leq 1 - c_0 \omega(t), \quad 0 < c_0 \leq \frac{1}{2}. \quad (7.1)$$

Here c_0 depends only on d . The shift may be fixed by conditioning on other variables as long as the selected component retains its law conditional on its attaching spin. This is the only quantitative information estimate used in the present section.

7.1 Products in both directions

For a vertex z and either the empty set or a singleton $F \subseteq \{u : u \sim z\}$, let $\mathcal{P}_\ell(z; F)$ consist of the simple paths $\gamma = (z_0, \dots, z_\ell)$ that start at z and, when $\ell \geq 1$, have $z_1 \notin F$. Thus

$$|\mathcal{P}_\ell(z; F)| = (d - |F|)b^{\ell-1} \quad (\ell \geq 1), \quad \mathcal{P}_0(z; F) = \{(z)\}.$$

Define the two squared products and their averages over paths by

$$\begin{aligned} R_\gamma^+ &= \prod_{i=1}^{\ell} a(h_{z_i \rightarrow z_{i-1}})^2, & G_\ell^+(z; F) &= \frac{1}{|\mathcal{P}_\ell(z; F)|} \sum_{\gamma \in \mathcal{P}_\ell(z; F)} R_\gamma^+, \\ R_\gamma^- &= \prod_{i=1}^{\ell} a(h_{z_{i-1} \rightarrow z_i})^2, & G_\ell^-(z; F) &= \frac{1}{|\mathcal{P}_\ell(z; F)|} \sum_{\gamma \in \mathcal{P}_\ell(z; F)} R_\gamma^-. \end{aligned}$$

Empty products are one, so $G_0^+ = G_0^- = 1$. We call the products outward and inward according to the coefficient traversal: R^+ is traversed from z_0 to z_ℓ , and R^- from z_ℓ to z_0 . In each factor the field arrow names the component root and its excluded neighbor, and points opposite to that coefficient traversal.

We will split the path at an intermediate vertex and average its continuations inside disjoint descendant components. Outward continuation products already use only their descendant observations. An inward product can also see the observations above the split. The next lemma removes the one entering message that causes this dependence.

Lemma 7.1 (A continuation below a cut). *Let y and p be adjacent vertices, and root the component $\mathcal{T}_y = T_{y \rightarrow p}$ at y . Write $\text{Ch}(w)$ for the children of w in this component. For a descendant path $y = w_0, w_1, \dots, w_L$ of length $L \geq 1$, set $H_i = h_{w_i \rightarrow w_{i+1}}$ for $0 \leq i < L$ and define*

$$\begin{aligned} \tilde{H}_0 &= X_y + \sum_{u \in \text{Ch}(y) \setminus \{w_1\}} \psi(h_{u \rightarrow y}), \\ \tilde{H}_i &= X_{w_i} + \psi(\tilde{H}_{i-1}) + \sum_{u \in \text{Ch}(w_i) \setminus \{w_{i+1}\}} \psi(h_{u \rightarrow w_i}), \quad 1 \leq i < L. \end{aligned} \tag{7.2}$$

Each $\tilde{H}_i$ is a function only of the observations in $\mathcal{T}_y$. With

$$C_{\text{cut}} = \exp\left(\frac{4\beta}{1-\theta}\right),$$

the original and modified inward products obey, almost surely,

$$\left| \log \frac{\prod_{i=0}^{L-1} a(H_i)^2}{\prod_{i=0}^{L-1} a(\tilde{H}_i)^2} \right| \leq \frac{4\beta}{1-\theta}. \tag{7.3}$$

Moreover, conditional on the true spins of this path, both the outward product and the modified inward product satisfy

$$\begin{aligned} \mathbb{E} \left[\prod_{i=1}^L a(h_{w_i \rightarrow w_{i-1}})^2 \middle| \sigma_{w_0}, \dots, \sigma_{w_L} \right] &\leq e^{-c_0 \omega(t)L}, \\ \mathbb{E} \left[\prod_{i=0}^{L-1} a(\tilde{H}_i)^2 \middle| \sigma_{w_0}, \dots, \sigma_{w_L} \right] &\leq e^{-c_0 \omega(t)L}. \end{aligned} \tag{7.4}$$

The constants are independent of t , L , and the choice of path.

Proof. The recurrence deletes only the contribution from p at y and propagates that change forward; every field from a side child is unchanged. All these fields use components contained in $\mathcal{T}_y$, so the claimed locality follows by induction. The modified fields are used as functions of the descendant observations; no posterior interpretation for a truncated model is needed.

The cavity identity gives $H_0 - \tilde{H}_0 = \psi(h_{p \rightarrow y})$. At each subsequent step, only the predecessor message differs. Since $|\psi| \leq \beta$ and $|\psi'| \leq \theta$, induction yields

$$|H_i - \tilde{H}_i| \leq \beta \theta^i \quad (0 \leq i < L).$$

The bound $|(\log a)'| \leq 2$ from (5.9) now gives

$$\left| \log \frac{\prod_{i=0}^{L-1} a(H_i)^2}{\prod_{i=0}^{L-1} a(\tilde{H}_i)^2} \right| \leq 4 \sum_{i=0}^{L-1} |H_i - \tilde{H}_i| \leq 4\beta \sum_{i=0}^{L-1} \theta^i \leq \frac{4\beta}{1-\theta}.$$

This proves (7.3). In particular, obtaining component locality costs a multiplicative factor at most C_{cut} , independent of the length and time.

The outward inequality in (7.4) is the conditional path estimate of Lemma 3.4, applied to $w_0, \dots, w_L$. We prove the modified inward inequality by integrating intact side components from the far end back toward the cut.

At each target w_i , $0 \leq i < L$, choose one child off the path, using only the path geometry. There are $b - 1 = d - 2 \geq 1$ choices, including at w_0 after deleting its parent and excluding w_1 . The selected child components are disjoint and infinite; this also covers $d = 3$ and $L = 1$.

Condition on the true path spins. The selected component observation histories are independent of each other and of the remaining observation histories, and their laws depend only on the attaching path spins. Indeed, root the broadcast at w_0 and group each selected component's attaching-edge transition, internal transitions, and Brownian noises. These groups are mutually independent; once the path spins are specified, the observations outside the selected components use none of them. This factorization passes from finite cylinder events to the sigma-fields of the full observation histories. Thus fixing the other observation histories leaves each selected component with exactly the conditional law used in (7.1).

Integrate the selected components in the order $w_{L-1}, w_{L-2}, \dots, w_0$. In (7.2), $\tilde{H}_i$ is its selected side message plus a shift fixed by the other observations. Every field still present after its factor is removed has index less than i and does not use that message. Equation (7.1) therefore removes each squared factor at a cost at most $1 - c_0 \omega(t)$. The resulting bound is $(1 - c_0 \omega(t))^L \leq e^{-c_0 \omega(t)L}$, as required. $\square$

Proposition 7.2 (Concentration of path averages). *Set*

$$\eta = \min \left\{ \frac{c_0}{4}, \frac{\log b}{8} \right\}, \quad C_0 = \max \left\{ 2, \frac{4 \log 2}{c_0} \right\}, \quad \ell_*(t) = \left\lceil \frac{C_0}{\omega(t)} \right\rceil.$$

For every $t > 0$, every vertex z , every allowed first-step restriction F , either orientation $\varepsilon \in \{+, -\}$, and every integer $\ell \geq \ell_(t)$,*

$$\mathbb{P} \left(G_\ell^\varepsilon(z; F) > C_{\text{cut}} e^{-\eta \omega(t) \ell} \right) \leq \exp \left(-\frac{b^{\ell/4}}{2b} \right). \quad (7.5)$$

The constants depend only on d and are uniform in the time, vertex, first-step restriction, and orientation.

Proof. Root the tree at z and put $k = \lfloor \ell/2 \rfloor$ and $L = \ell - k$. The threshold ensures $k \geq 1$. Let $\mathcal{Y}_k$ be the vertices at depth k whose path from z obeys the restriction F . There are

$$N := |\mathcal{Y}_k| = (d - |F|) b^{k-1} \geq b^k \quad (7.6)$$

such vertices, and each has exactly b^L descendant continuations of length L . Since every squared factor lies in $(0, 1]$, deleting the first k factors increases either product. The average of the remaining products is consequently an equal-weight average of N continuation averages.

For $y \in \mathcal{Y}_k$, let $p(y)$ be its parent and $\mathcal{T}_y = T_{y \rightarrow p(y)}$. Let Z_y^+ be the average of

$$\prod_{i=1}^L a(h_{w_i \rightarrow w_{i-1}})^2$$

over its b^L continuations $y = w_0, \dots, w_L$. Let Z_y^- be the average of $\prod_{i=0}^{L-1} a(\tilde{H}_i)^2$ for the same continuations, using (7.2) separately for each one. Both variables lie in $[0, 1]$ and are functions only of the observation histories in $\mathcal{T}_y$. Deleting the first k factors and applying (7.3) gives

$$G_\ell^\varepsilon(z; F) \leq \frac{C_{\text{cut}}}{N} \sum_{y \in \mathcal{Y}_k} Z_y^\varepsilon, \quad \varepsilon \in \{+, -\}. \quad (7.7)$$

For $+$ the multiplier can be replaced by one.

Averaging (7.4) first over the continuation spins conditional on σ_y , and then over the continuation choices, yields

$$\mathbb{E}[Z_y^\varepsilon \mid \sigma_y] \leq e^{-c_0 \omega(t)L}, \quad \varepsilon \in \{+, -\}. \quad (7.8)$$

Now condition on the sigma-field of true cut-level spins

$$\mathcal{H}_k = \sigma(\sigma_y : y \in \mathcal{Y}_k).$$

The observations in the disjoint components $\mathcal{T}_y$ are independent under this conditioning, with each component law depending only on its own root spin. Indeed, after specifying the cut-level spins, all broadcast transitions and Brownian noises strictly below the cut, together with the Brownian noise at each cut vertex, can be sampled independently across these components. The same statement holds when F forbids one first neighbor, since that simply omits one collection of components. The cut-level spins themselves need not be independent. Thus the Z_y^ε are conditionally independent variables in $[0, 1]$, and their conditional means given $\mathcal{H}_k$ satisfy (7.8).

Write $\tau_\ell = e^{-\eta \omega(t)\ell}$. Since $L \geq \ell/2$, $\omega(t)\ell \geq C_0$, and $\eta \leq c_0/4$, the chosen constants give

$$\mathbb{E}[Z_y^\varepsilon \mid \mathcal{H}_k] \leq e^{-c_0 \omega(t)\ell/2} \leq \frac{1}{2} \tau_\ell.$$

Apply Hoeffding's bounded-sum inequality [8, Theorem 2] to each conditional product law. Because the distance from the conditional mean of the average to τ_ℓ is at least $\tau_\ell/2$, it gives

$$\begin{aligned} \mathbb{P}\left(\frac{1}{N} \sum_y Z_y^\varepsilon > \tau_\ell \mid \mathcal{H}_k\right) &\leq \exp\left(-2N(\tau_\ell/2)^2\right) \\ &= \exp\left(-\frac{1}{2} N e^{-2\eta \omega(t)\ell}\right) \leq \exp\left(-\frac{b^{\ell/4}}{2b}\right). \end{aligned}$$

For the last inequality, $k \geq \ell/2 - 1$, $\omega(t) \leq 1$, and $2\eta \leq (\log b)/4$ imply

$$N e^{-2\eta \omega(t)\ell} \geq b^{\ell/2-1} e^{-(\log b)\ell/4} = b^{\ell/4}/b.$$

The conditional bound holds almost surely. Taking expectations and using (7.7) proves (7.5). $\square$

Here is the consequence that makes concentration useful for dependent weights. Let ζ be any bounded nonnegative random variable on a probability space carrying an observation array with the planted marginal law. It may also depend on further variables coupled to that array. For $\ell \geq \ell_*(t)$, split according to the event in (7.5); on its complement use the displayed bound for G_ℓ^ε , and on the event use $G_\ell^\varepsilon \leq 1$. Then

$$\mathbb{E}[\zeta G_\ell^\varepsilon(z; F)] \leq C_{\text{cut}} e^{-\eta \omega(t) \ell} \mathbb{E} \zeta + \|\zeta\|_\infty \varepsilon_\ell, \quad \varepsilon_\ell := \exp\left(-\frac{b^{\ell/4}}{2b}\right). \quad (7.9)$$

Only the observation marginal enters the event probability. The principal term therefore pays the mean of the weight even when the weight depends on all those observations.

7.2 Averaging with the discrepancy weights

Return to the invariant joined law of the two observation copies. Recall from (5.11) that, on a directed path $v_0, \dots, v_r$ with $e_i = (v_{i-1}, v_i)$,

$$P_r = \prod_{i=1}^r a(h_{v_i \rightarrow v_{i-1}}), \quad Q_r = \mathbb{E} \left[(P_r^2 + (P'_r)^2) \left(\sum_{i=1}^r D_{e_i} \right)^2 \right], \quad L_r = \mathbb{E}[(\Delta h_{v_0})^2 P_r^2].$$

Here $D_{(i,j)} = \log a(h_{j \rightarrow i}) - \log a(h'_{j \rightarrow i})$, $S = \mathbb{E} D_e^2$, and $V = \mathbb{E}(\Delta h_v)^2$. Invariance makes S, V, Q_r, L_r independent of the chosen edge, vertex, or directed path of that length. The path averages below use that invariance and leave all observation fields unconditioned.

Proposition 7.3 (Weighted path bounds). *For every finite $T > 0$ there are $c, C > 0$, depending only on d , and $C_T < \infty$, depending also on T , such that for $0 < t \leq T$ and every integer $r \geq 2\ell_*(t)$,*

$$\begin{aligned} Q_r(t) &\leq C e^{-c\omega(t)r} r S(t) + C_T r^2 \exp(-cb^{cr}), \\ L_r(t) &\leq C e^{-c\omega(t)r} V(t) + C_T \exp(-cb^{cr}). \end{aligned} \quad (7.10)$$

At every integer $r \geq 1$, without a lower length restriction,

$$Q_r(t) \leq 2C_{\text{var}} r S(t), \quad L_r(t) \leq V(t), \quad (7.11)$$

where C_{var} is the constant in Proposition 6.1. The hypothesis for (7.10) follows in particular from $r \geq 2(C_0 + 1)/\omega(t)$.

Proof. Fix T and put

$$H_T = 2T + 2d\beta, \quad D_{\max, T} = \frac{2H_T}{1-\theta}.$$

The site bound (5.5), the component comparison (5.8), and $|(\log a)'| \leq 2$ imply

$$|\Delta h_v| \leq H_T, \quad |\Delta h_{j \rightarrow i}| \leq \frac{H_T + \theta H_T}{1 - \theta^2} = \frac{H_T}{1 - \theta}, \quad |D_e| \leq D_{\max, T}. \quad (7.12)$$

Dropping $P_r^2, (P'_r)^2 \leq 1$ and applying Proposition 6.1 gives the first bound in (7.11); dropping P_r^2 gives the second.

For the longer paths, set

$$p = \lfloor r/2 \rfloor, \quad q = r - p, \quad U = \sum_{i=1}^p D_{(v_{i-1}, v_i)}, \quad Z = \sum_{i=p+1}^r D_{(v_{i-1}, v_i)}.$$

Since $(U + Z)^2 \leq 2U^2 + 2Z^2$,

$$Q_r \leq 2\mathbb{E}[(P_r^2 + (P'_r)^2)U^2] + 2\mathbb{E}[(P_r^2 + (P'_r)^2)Z^2]. \quad (7.13)$$

Figure 1 displays the two path averages that control these terms.

First fix the prefix $(v_0, \dots, v_p)$ and vary all length- q extensions from v_p that avoid v_{p-1} on their first step. The pointwise stabilizer of the prefix is transitive on these extensions: map the successive descendant branches of one extension to those of the other, and extend independently to the remaining regular rooted subtrees. The invariance of the joined law consequently makes the expectation of the product for each complete path, multiplied by U^2 , the same for every extension. In a fixed joined observation configuration, U also denotes the same quantity in every term of this finite average: its indexed directed edges stay fixed. This statement places no locality restriction on their fields. Dropping the prefix factors from the product gives

$$\mathbb{E}[P_r^2 U^2] \leq \mathbb{E}[U^2 G_q^+(v_p; \{v_{p-1}\})]. \quad (7.14)$$

For the other term, fix the suffix $(v_p, \dots, v_r)$ and vary a reversed prefix $w_0 = v_p, w_1, \dots, w_p$ with $w_1 \neq v_{p+1}$. The resulting complete directed path is

$$(w_p, w_{p-1}, \dots, w_1, v_p, v_{p+1}, \dots, v_r).$$

The pointwise stabilizer of the suffix is transitive on these choices; it may move the original starting vertex v_0 , which invariance under the full automorphism group permits. The suffix sum Z has the same indexed edges in every term. In the direction of the complete path, its prefix product is exactly

$$\prod_{i=1}^p a(h_{w_{i-1} \rightarrow w_i})^2,$$

the inward product along the reversed geometric prefix. Dropping the suffix factors therefore yields

$$\mathbb{E}[P_r^2 Z^2] \leq \mathbb{E}[Z^2 G_p^-(v_p; \{v_{p+1}\})]. \quad (7.15)$$

Each path average is finite. Thus its use of equivariance requires only a finite intersection of almost-sure identities for fixed automorphisms, consistent with the theorem's convention.

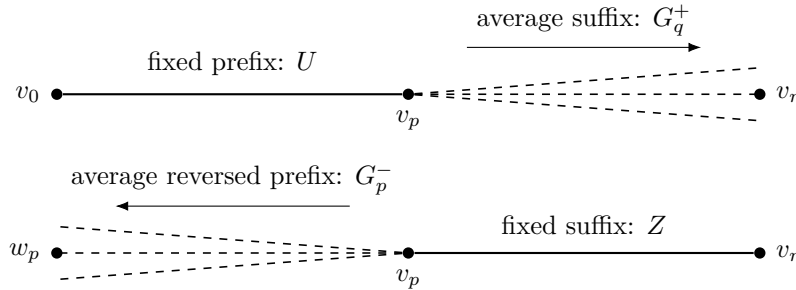


Figure 1: The two averages in the half-path argument. Solid halves are fixed and dashed halves are varied; each segment represents a half-path, with one representative pair of endpoints labeled. Arrows show the direction in which geometric extensions are enumerated. In the lower row, the prefix is enumerated from v_p toward w_p , while its coefficients are traversed from w_p to v_p , giving the inward average G_p^- . The observation fields remain unconditioned.

Suppose $r \geq 2\ell_*(t)$. Then $p, q \geq \ell_*(t)$. Apply (7.9) to the right side of (7.14) with $\zeta = U^2$. The unconditional path variance bound and (7.12) give $\mathbb{E}U^2 \leq C_{\text{var}}pS$ and $\|U^2\|_\infty \leq p^2D_{\text{max},T}^2$. Hence

$$\mathbb{E}[P_r^2U^2] \leq C_{\text{cut}}C_{\text{var}}e^{-\eta\omega(t)q}pS + D_{\text{max},T}^2p^2\varepsilon_q. \quad (7.16)$$

Applying the same bound to (7.15), now with $\zeta = Z^2$, gives

$$\mathbb{E}[P_r^2Z^2] \leq C_{\text{cut}}C_{\text{var}}e^{-\eta\omega(t)p}qS + D_{\text{max},T}^2q^2\varepsilon_p. \quad (7.17)$$

These applications use the probability, under the joined law, of the corresponding event determined by one observation marginal. They do not require a conditional variance bound or independence of that event from the discrepancy. The identical reasoning applies to $(P'_r)^2$ using the second marginal's path averages.

For $r \geq 2$, both p and q are at least $r/3$. Inserting the four estimates into (7.13), and using $p + q = r$ and $p^2 + q^2 \leq r^2$, proves the explicit inequality

$$Q_r \leq 4C_{\text{cut}}C_{\text{var}}e^{-\eta\omega(t)r/3}rS + 4D_{\text{max},T}^2r^2 \exp\left(-\frac{br^{1/2}}{2b}\right). \quad (7.18)$$

Finally fix v_0 and average the entire directed path over $\mathcal{P}_r(v_0; \emptyset)$. The stabilizer of v_0 is transitive on these paths, and the vertex quantity Δh_{v_0} is fixed in every term. Invariance gives the exact identity

$$L_r = \mathbb{E}[(\Delta h_{v_0})^2 G_r^+(v_0; \emptyset)].$$

For $r \geq \ell_*(t)$, use (7.9) with $\zeta = (\Delta h_{v_0})^2$, whose mean is V and whose essential supremum is at most H_T^2 . This yields

$$L_r \leq C_{\text{cut}}e^{-\eta\omega(t)r}V + H_T^2 \exp\left(-\frac{br^{1/2}}{2b}\right). \quad (7.19)$$

Choosing one sufficiently small positive c in the two explicit bounds gives (7.10). Finally, $\ell_*(t) \leq (C_0 + 1)/\omega(t)$ since $\omega(t) \leq 1$, which proves the stated sufficient length condition. $\square$

8 Critical comparison and recovery

We now show that two conditional copies with the same innovations agree at the critical parameter $\theta = b^{-1/2}$, where $b = d - 1 \geq 2$. The weighted path estimates reduce the question to the variance $V(t) = \mathbb{E}[(\Delta h_o(t))^2]$ of the site-field difference. Their principal noise term has the scale

$$S(t) \sum_{r \geq 1} r e^{-c\sqrt{t}r} \asymp \frac{S(t)}{t} \quad (t \downarrow 0),$$

where $S(t) = \mathbb{E}[D_e(t)^2]$ and $D_{(i,j)} = \log a(h_{j \rightarrow i}) - \log a(h'_{j \rightarrow i})$. The earlier estimate $S \leq CV$ would leave a coefficient $1/t$, which is not integrable at zero. We improve it by using the fact that $(\log a)'$ vanishes at the origin. A field is usually small at early times; the first result below makes the exceptional probability small enough to compare it with any positive value of V .

8.1 All moments and a small-field tail

We use the fixed-time posterior identities and critical information estimate already proved for one observation process. At a deterministic time $t > 0$, fix an oriented edge $v \rightarrow w$ and write

$$Y = h_{v \rightarrow w}(t), \quad M(t) = \mathbb{E} \tanh^2 Y.$$

The law of Y is independent of the choice of edge and is symmetric. Recall that $M(t) \leq C\sqrt{t}$ for $0 < t \leq 1$, that $|Y - X_v(t)| \leq b\beta$, and that $X_v(t) = t\sigma_v + \sqrt{t}G$ with G a standard normal variable independent of the spin. These are statements about the planted observation law. Each marginal of the conditional coupling has this same law.

The second moment is critical because $b\theta^2 = 1$, whereas $b\theta^p < 1$ for every integer $p \geq 3$; this strict contraction lets the recursion propagate the second-moment scale to every higher integer order.

Lemma 8.1 (Critical field moments and tail). *There are constants $0 < K < \infty$ and $t_0 \in (0, 1)$, depending only on d , such that for every integer $p \geq 1$ and every $0 < t \leq t_0$,*

$$\mathbb{E}|h_{v \rightarrow w}(t)|^p \leq e^{Kp^2} t^{p/4}. \quad (8.1)$$

There are also $c > 0$ and $t_{\text{tail}} \in (0, t_0]$ such that

$$\mathbb{P}(|h_{v \rightarrow w}(t)| > t^{1/8}) \leq \exp(-c \log^2(1/t)), \quad 0 < t \leq t_{\text{tail}}. \quad (8.2)$$

Both inequalities hold as well when conditioned on either value of the component-root spin σ_v , or on either value of the deleted-neighbor spin σ_w .

Proof. We begin with the second moment. On $\{|X_v(t)| \leq 1\}$ the cavity bound gives $|Y| \leq 1 + b\beta$. Comparison on this compact interval therefore gives $Y^2 \leq C \tanh^2 Y$. On its complement we use $Y^2 \leq 2X_v(t)^2 + 2b^2\beta^2$. For $t \leq 1/2$, the event $|t\sigma_v + \sqrt{t}G| > 1$ requires $|G| > 1/(2\sqrt{t})$. Integration of the normal density, decreasing the positive exponent constant if necessary, yields

$$\mathbb{E}[(1 + X_v(t)^2)\mathbf{1}_{\{|X_v(t)| > 1\}}] \leq Ce^{-c/t}.$$

Combining the two events with $M(t) \leq C\sqrt{t}$ proves that, after fixing some $t_0 \leq 1/2$,

$$A_2(t) := \mathbb{E}|Y|^2 \leq C\sqrt{t}, \quad A_1(t) := \mathbb{E}|Y| \leq C^{1/2}t^{1/4} \quad (0 < t \leq t_0). \quad (8.3)$$

The second inequality is Cauchy–Schwarz. The cavity bound also shows that $A_p(t) := \mathbb{E}|Y|^p$ is finite for every integer $p \geq 1$.

We next explain why the recursion can be expanded using independent terms without changing these absolute moments. For an integrable function f , the spin-tilt identities from the posterior calculation are

$$\begin{aligned} \mathbb{E}[f(Y) \mid \sigma_v = \eta] &= \mathbb{E}[f(Y)(1 + \eta \tanh Y)], \\ \mathbb{E}[f(Y) \mid \sigma_w = \eta] &= \mathbb{E}[f(Y)(1 + \eta\theta \tanh Y)], \quad \eta \in \{-1, 1\}. \end{aligned}$$

These are the identities (3.6) from Lemma 3.3. Because Y has a symmetric marginal law, the term containing $\tanh Y$ integrates to zero against every even function of Y . In particular conditioning on either spin leaves each $A_p(t)$ unchanged.

Condition now on $\sigma_v = +1$ in the cavity recursion at v . The own observation and the b child-component observations are independent under this conditional law: their spin transitions and their

planted noises lie in disjoint branches given the root spin. The own observation has law $t + \sqrt{t} G$. A child field is conditioned through its deleted-neighbor spin, so the even-function consequence of the second tilt gives, for every integer $r \geq 1$,

$$\mathbb{E}[|\psi(h_{u \rightarrow v}(t))|^r \mid \sigma_v = +1] = \mathbb{E}|\psi(Y)|^r \leq \theta^r A_r(t). \quad (8.4)$$

Here $\psi(0) = 0$ and $0 < \psi'(z) \leq \theta$ imply $|\psi(z)| \leq \theta|z|$.

There is an absolute constant K_0 such that

$$\mathbb{E}|t + \sqrt{t} G|^r \leq e^{K_0 r^2 t^{r/4}} \quad (r \geq 1 \text{ an integer}, 0 < t \leq 1). \quad (8.5)$$

Indeed, $|t + \sqrt{t} G| \leq \sqrt{t}(1 + |G|)$ in this time range, and $\mathbb{E}|G|^r \leq (C\sqrt{r})^r$. The inequality $(x + y)^r \leq 2^{r-1}(x^r + y^r)$ for nonnegative x, y bounds the remaining constant by $e^{K_0 r^2}$, uniformly over integer r . We also used $t^{r/2} \leq t^{r/4}$.

Choose $K \geq 2K_0$ large enough that (8.1) holds for $p = 1, 2$, using (8.3). Suppose it holds for all positive integers below p , where $p \geq 3$. The triangle inequality in the cavity recursion, followed by the multinomial expansion and the conditional independence just established, bounds $A_p(t)$ by the expectation of the p th power of the sum of the absolute values of the own term and the b child terms. The terms with exponent p on a single child contribute at most $b\theta^p A_p(t)$. The term supported on the own observation is at most $e^{Kp^2/2} t^{p/4}$ by (8.5).

Every other exponent vector $(r_0, r_1, \dots, r_b)$ has sum p and at least two positive entries. Hence every positive entry is below p , and the induction hypothesis applies to its child factors. Using (8.5) for its own factor, if present, and discarding powers of $\theta \leq 1$, its product of moments is at most

$$e^{K \sum_{i=0}^b r_i^2} t^{p/4}.$$

For such a vector, $\sum_i r_i^2 \leq (p-1)^2 + 1 = p^2 - 2(p-1)$: merging positive entries until only two remain, then moving units to the larger one, can only increase the sum of squares. The sum of all multinomial coefficients in $b+1$ slots is $(b+1)^p$. Thus

$$(1 - b\theta^p) A_p(t) \leq e^{Kp^2} t^{p/4} \left[e^{-Kp^2/2} + (b+1)^p e^{-2K(p-1)} \right]. \quad (8.6)$$

The coefficient on the left is positive uniformly over $p \geq 3$:

$$1 - b\theta^p = 1 - b^{1-p/2} \geq \kappa_b := 1 - b^{-1/2} > 0.$$

Enlarge the same constant K once more so that

$$2K > \log(b+1), \quad e^{-9K/2} + (b+1)^3 e^{-4K} \leq \kappa_b.$$

Both terms in brackets in (8.6) decrease as the integer p increases from 3. Their sum is therefore at most $\kappa_b \leq 1 - b\theta^p$ for every such p . Absorption proves (8.1) with this single K , for every integer order and all $t \leq t_0$.

To obtain the tail, put $L = \log(1/t)$ and choose the integer $p = \lfloor L/(16K) \rfloor$. If $L \geq 32K$, then $L/(32K) \leq p \leq L/(16K)$. Markov's inequality gives

$$\mathbb{P}(|Y| > t^{1/8}) \leq e^{Kp^2 - pL/8} \leq e^{-pL/16} \leq e^{-L^2/(512K)}.$$

Take $t_{\text{tail}} \leq \min(t_0, e^{-32K})$. Both $|Y|^p$ and the indicator of the tail event are even functions of Y , so the spin-tilt identities prove all the conditional assertions as well. $\square$

8.2 Comparing log-factor and site-field differences

Fix a finite horizon $T > 0$. The conditional-copy construction gives

$$|\Delta h_v(t)| \leq H_T := 2T + 2d\beta, \quad 0 \leq V(t) \leq H_T^2 \quad (0 \leq t \leq T),$$

and Lemma 5.2 makes V absolutely continuous, with $V(0) = 0$. Choose a constant, fixed throughout this horizon,

$$K_T > \max\{1, H_T^2\}, \quad I(t) = 1 + \log \frac{K_T}{V(t)} \quad \text{when } V(t) > 0. \quad (8.7)$$

In particular $I(t) > 1$, including at times when $V(t) \geq 1$.

Lemma 8.2 (Small-time log-factor estimate). *There is a fixed $t_* \in (0, \min\{t_{\text{tail}}, e^{-1}\}]$ such that, for $0 < t \leq \min\{t_*, T\}$ with $V(t) > 0$,*

$$S(t) \leq C_T V(t) \left(t^{1/4} + \frac{I(t)}{\log^2(1/t)} \right). \quad (8.8)$$

At every time in $[0, T]$, the bound $S(t) \leq CV(t)$ remains valid.

Proof. Write $Y = h_{j \rightarrow i}(t)$ and $Y' = h'_{j \rightarrow i}(t)$ for the edge used to define $S(t)$. The two cavity equations on this edge and $|\psi'| \leq \theta$ give the earlier comparison

$$|Y - Y'| \leq \frac{|\Delta h_j| + \theta |\Delta h_i|}{1 - \theta^2}.$$

Invariance and Cauchy–Schwarz consequently give

$$\mathbb{E}|Y - Y'|^2 \leq \frac{V(t)}{(1 - \theta)^2}, \quad |Y - Y'| \leq \frac{H_T}{1 - \theta}. \quad (8.9)$$

Also

$$|(\log a)'(z)| = \frac{2(1 - \theta^2)|\tanh z|}{1 - \theta^2 \tanh^2 z} \leq 2|\tanh z| \leq 2.$$

The global bound gives $S(t) \leq 4V(t)/(1 - \theta)^2$ and $|D_e(t)| \leq 2H_T/(1 - \theta)$.

On the event $|Y|, |Y'| \leq t^{1/8}$ the segment between them also lies in that interval, and $|(\log a)'(z)| \leq 2t^{1/8}$ there. The mean-value theorem and (8.9) therefore give

$$\mathbb{E}[D_e(t)^2; |Y|, |Y'| \leq t^{1/8}] \leq \frac{4}{(1 - \theta)^2} t^{1/4} V(t).$$

For the complementary event the global square-integrable bound gives $CV(t)$, while the deterministic bound on D_e and a union bound using (8.2) in the two marginals give $C_T e^{-c \log^2(1/t)}$. Thus

$$\mathbb{E}[D_e(t)^2; \max(|Y|, |Y'|) > t^{1/8}] \leq \min\{CV(t), C_T e^{-c \log^2(1/t)}\}. \quad (8.10)$$

The argument uses only the two marginal probabilities; it requires no independence between the two fields.

Set $u = V(t) > 0$ and $J = \log^2(1/t) \geq 1$. If $u \leq e^{-cJ/2}$, then $I(t) \geq cJ/2$, so the first member of the minimum in (8.10) is at most $CuI(t)/J$. If $u > e^{-cJ/2}$, the second member divided by u is at most $C_T e^{-cJ/2}$. Since $Je^{-cJ/2}$ is bounded for $J \geq 1$ and $I(t) \geq 1$, this is at most $C_T I(t)/J$. Adding the small-field contribution proves (8.8). $\square$

8.3 A cutoff depending on the discrepancy

We next sum the weighted estimates. Recall $\omega(t) = \min\{\sqrt{t}, 1\}$, and suppress t at a fixed time with $0 < t \leq T$ and $V(t) > 0$. The preceding path results give

$$\begin{aligned} Q_r &\leq CrS, & L_r &\leq V & (r \geq 1), \\ Q_r &\leq Ce^{-c\omega r}rS + C_T r^2 e^{-\alpha b^{\gamma r}} & (r \geq C'_0/\omega), \\ L_r &\leq Ce^{-c\omega r}V + C_T e^{-\alpha b^{\gamma r}} & (r \geq C'_0/\omega), \end{aligned}$$

for fixed positive constants c, α, γ, C'_0 depending only on d . The last two lines follow from Proposition 7.3, with constants decreased or enlarged so that both exceptional terms have the indicated common form. These estimates include weights and discrepancies formed from the same observations.

Choose C_1 sufficiently large that

$$C_1 \geq C'_0, \quad \gamma C_1 \log b \geq 1, \quad (\alpha/2)b^{\gamma C_1} \geq 2,$$

and take the integer

$$R = \lceil C_1(\omega^{-1} + \log I) \rceil. \quad (8.11)$$

The weighted estimates apply for $r > R$. Since $\omega^{-1} \geq 1$ and $I > 1$, the choices above also give

$$(\alpha/2)b^{\gamma r} \geq (\alpha/2)b^{\gamma C_1 \omega^{-1}} I^{\gamma C_1 \log b} \geq 2I \quad (r > R).$$

Splitting the exceptional exponent into equal halves therefore gives

$$e^{-\alpha b^{\gamma r}} \leq e^{-2I} e^{-(\alpha/2)b^{\gamma r}} = e^{-2} \left(\frac{V}{K_T} \right)^2 e^{-(\alpha/2)b^{\gamma r}} \leq V^2 e^{-(\alpha/2)b^{\gamma r}} \quad (r > R). \quad (8.12)$$

The last step uses $K_T > 1$, and so remains valid when $V \geq 1$.

Lemma 8.3 (The noise and drift path sums). *At each time $0 < t \leq T$ with $V(t) > 0$,*

$$\begin{aligned} \sum_{r \geq 1} Q_r &\leq C_T \left[S(\omega^{-2} + (\log I)^2) + V^2 \right], \\ \sum_{r \geq 1} \sqrt{Q_r} &\leq C_T \left[\sqrt{S}(\omega^{-3/2} + (\log I)^{3/2}) + V \right], \\ \sum_{r \geq 1} L_r &\leq C_T \left[V(\omega^{-1} + \log I) + V^2 \right]. \end{aligned} \quad (8.13)$$

Proof. For $r \leq R$, the estimates $Q_r \leq CrS$ and $L_r \leq V$ give the respective bounds

$$CS \sum_{r \leq R} r \leq CSR^2, \quad C\sqrt{S} \sum_{r \leq R} \sqrt{r} \leq C\sqrt{S}R^{3/2}, \quad \sum_{r \leq R} L_r \leq VR.$$

For the ordinary exponential terms at $r > R$, use

$$\sum_{r \geq 1} r^p e^{-c\omega r} \leq C_{p,c} \omega^{-p-1} \quad \left(p \in \left\{ 0, \frac{1}{2}, 1 \right\}, \quad 0 < \omega \leq 1 \right). \quad (8.14)$$

For example, compare the sum with $\int_0^\infty (x+1)^p e^{-c\omega x} dx$ and put $y = \omega x$; since $\omega \leq 1$, the resulting integral is bounded by $\omega^{-p-1} \int_0^\infty (y+1)^p e^{-cy} dy$. This gives $CS\omega^{-2}$ for the first sum and $CV\omega^{-1}$

for the third. For the middle sum use $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$; the ordinary exponential contribution is bounded by $C\sqrt{S} \sum_{r \geq 1} r^{1/2} e^{-\omega r/2} \leq C\sqrt{S} \omega^{-3/2}$.

For the exceptional terms, (8.12) bounds their three contributions by constants times

$$V^2 \sum_{r \geq 1} r^2 e^{-(\alpha/2)b^{\gamma r}}, \quad V \sum_{r \geq 1} r e^{-(\alpha/4)b^{\gamma r}}, \quad V^2 \sum_{r \geq 1} e^{-(\alpha/2)b^{\gamma r}},$$

respectively. Each series is finite because its exponential exponent grows exponentially in r . In the middle series the square root of the V^2 factor is V , which is why the drift sum has this order. Finally $\omega^{-1} + \log I \geq 1$, so the ceiling in (8.11) gives $R \leq (C_1 + 1)(\omega^{-1} + \log I)$. Bounding each power of this sum by a constant times the sum of its corresponding powers proves (8.13). $\square$

8.4 An integrable comparison at time zero

Proposition 8.4 (Critical conditional-copy uniqueness). *For the invariant conditional coupling with common innovations at $\theta = b^{-1/2}$, the site-field discrepancy satisfies $V(t) = 0$ for every $t \geq 0$.*

Proof. Work on the fixed horizon $[0, T]$ used above. Inserting the noise and drift bounds (5.13) and (5.14) into the variance identity (5.6), and using $n_r \theta^{2r} = db^{r-1} b^{-r} = d/b$ for $r \geq 1$, gives

$$V'(t) \leq C \left(V(t) + \sum_{r \geq 1} Q_r(t) + \sum_{r \geq 1} L_r(t) + \sqrt{V(t)} \sum_{r \geq 1} \sqrt{Q_r(t)} \right) \quad (8.15)$$

for almost every t . The sums in Lemma 8.3 are finite at the positive times with $V(t) > 0$ where we now use this inequality.

Insert those three bounds into (8.15). Use $S \leq CV$, and hence $\sqrt{VS} \leq CV$, for all terms except $S\omega^{-2}$. The bounds $V \leq H_T^2$ imply $V^{3/2} \leq H_T V$ and $V^2 \leq H_T^2 V$. Finally

$$(\log I)^p \leq C_p I \quad (I \geq 1, 0 < p \leq 2),$$

since $x^p e^{-x}$ is bounded on $x \geq 0$. As $\omega^{-1} \leq \omega^{-3/2}$, all the terms just treated are at most

$$C_T V(I + \omega^{-3/2}).$$

This identifies the worst drift contribution explicitly: $\sqrt{VS} \omega^{-3/2} \leq CV \omega^{-3/2}$.

For $0 < t \leq \min\{t_*, T\}$, we have $\omega = \sqrt{t}$. Lemma 8.2 gives the remaining estimate

$$S\omega^{-2} \leq C_T V \left(t^{-3/4} + \frac{I}{t \log^2(1/t)} \right).$$

Together with $I \geq 1$, the preceding bounds prove

$$V'(t) \leq k_T(t) V(t) \left(1 + \log \frac{K_T}{V(t)} \right) \quad \text{for almost every } t \in (0, T] \text{ with } V(t) > 0, \quad (8.16)$$

where, after increasing C_T , we may take

$$k_T(t) = \begin{cases} C_T \left(1 + t^{-3/4} + \frac{1}{t \log^2(1/t)} \right), & 0 < t \leq \min\{t_*, T\}, \\ C_T, & t_* < t \leq T. \end{cases} \quad (8.17)$$

For the second branch, $\omega(t) \geq \sqrt{t_*} > 0$ because $t_* \leq 1$; there we use $S \leq CV$ also for $S\omega^{-2}$. Thus that branch is bounded on the whole later interval. If $T \leq t_*$ there is no later interval. The small-time expression is used only for $t \leq t_* \leq e^{-1}$, and it is integrable there:

$$\int_0^{t_*} t^{-3/4} dt = 4t_*^{1/4}, \quad \int_0^{t_*} \frac{dt}{t \log^2(1/t)} = \frac{1}{\log(1/t_*)}.$$

In particular $k_T \in L^1(0, T)$.

We finish with an Osgood-type comparison [16], proving the needed absolutely continuous form directly. Suppose that V is positive somewhere on $[0, T]$. Choose such a time t , and let $a < t$ be the left endpoint of its component in the positivity set of V . Continuity and $V(0) = 0$ give $V(a) = 0$. On any closed interval $[u, t]$ with $a < u < t$, continuity bounds V away from zero. The function $z \mapsto \log(1 + \log(K_T/z))$ has bounded derivative on the range of V on that interval. Its composition with the absolutely continuous function V is therefore absolutely continuous, and the chain rule and (8.16) give

$$(\log I)' = -\frac{V'}{VI} \geq -k_T \quad \text{almost everywhere on } [u, t].$$

Integrating yields

$$\log I(u) \leq \log I(t) + \int_u^t k_T(s) ds \leq \log I(t) + \int_a^t k_T(s) ds < \infty.$$

The last bound is independent of u . But $V(u) \rightarrow V(a) = 0$ as $u \downarrow a$, so $I(u) \rightarrow \infty$ and $\log I(u) \rightarrow \infty$, a contradiction. Hence V vanishes on $[0, T]$. Since the horizon was arbitrary, it vanishes at every time. $\square$

8.5 Recovering the observations and the spin factor

The comparison has now proved $V(t) = 0$ for the particular copies sampled in (5.1): Proposition 5.3 handles $b\theta^2 < 1$, and Proposition 8.4 handles $b\theta^2 = 1$. We first deduce equality of their observation paths and turn it into a Borel map from innovations to observations. The spins will then be read from the integer-time slopes of those paths.

The final passage to a spin map equivariant on every input uses local conditional expectations.

Lemma 8.5 (Pointwise equivariance of spin factors). *Let $(U_v)_{v \in \mathcal{V}}$ be independent uniform labels on T_d . Suppose a Borel map $\Phi : [0, 1]^\mathcal{V} \rightarrow \{-1, 1\}^\mathcal{V}$ satisfies $\Phi(gU) = g\Phi(U)$ almost surely for each fixed $g \in \text{Aut}(T_d)$. There is a Borel map $\Psi : [0, 1]^\mathcal{V} \rightarrow \{-1, 1\}^\mathcal{V}$ such that $\Psi(U) = \Phi(U)$ almost surely and*

$$\Psi(gx) = g\Psi(x) \quad \text{for every } x \in [0, 1]^\mathcal{V} \text{ and } g \in \text{Aut}(T_d).$$

Proof. Fix an auxiliary vertex o . For each integer $R \geq 0$, choose a Borel function $h_R : [0, 1]^{B_R(o)} \rightarrow [-1, 1]$ representing

$$\mathbb{E}[\Phi(U)_o \mid U|_{B_R(o)}].$$

A Borel version exists, and clipping it to $[-1, 1]$ preserves this conditional expectation. Let G_R be the finite automorphism group of the rooted ball $B_R(o)$, and average:

$$q_R(y) = \frac{1}{|G_R|} \sum_{a \in G_R} h_R(ay).$$

Every $a \in G_R$ extends to a tree automorphism fixing o : outside the ball, match the isomorphic rooted components attached to its boundary. Such an extension preserves the IID law and the ball's

label sigma-field. The assumed almost-sure equivariance therefore makes $h_R(a(U|_{B_R(o)}))$ another version of the same conditional expectation. Thus q_R still represents that conditional expectation, and is Borel, bounded by one, and invariant under G_R on every input.

Apply this same local rule at every vertex. Explicitly, for a rooted isomorphism $\iota : B_R(o) \rightarrow B_R(v)$, put

$$A_R(x)_v = q_R((x_{\iota(u)})_{u \in B_R(o)}).$$

Two choices of ι differ by an element of G_R , so the value does not depend on this choice. The maps A_R are consequently Borel and satisfy $A_R(gx) = gA_R(x)$ for every input and automorphism. Each ι also extends to a tree automorphism taking o to v . Transporting the conditional expectation by this fixed automorphism shows that

$$A_R(U)_v = \mathbb{E}[\Phi(U)_v \mid U|_{B_R(v)}] \quad \text{almost surely.}$$

The ball sigma-fields increase to the full label sigma-field. Upward martingale convergence, together with a countable intersection over vertices and radii, gives $A_R(U)_v \rightarrow \Phi(U)_v$ simultaneously at every vertex almost surely.

On every label input define

$$\Psi(x)_v = \begin{cases} 1, & \limsup_{R \rightarrow \infty} A_R(x)_v \geq 0, \\ -1, & \limsup_{R \rightarrow \infty} A_R(x)_v < 0. \end{cases}$$

Boundedness makes each limsup finite, and this formula defines a Borel spin map. The pointwise equivariance of every A_R passes through the limsup and the common threshold. On the simultaneous convergence event the limits belong to $\{-1, 1\}$, so $\Psi(U) = \Phi(U)$ almost surely. $\square$

Proposition 8.6 (Recovery of observations and spins). *Fix $d \geq 3$ and $0 < \theta = \tanh \beta \leq (d-1)^{-1/2}$, and retain the path spaces and laws $\mathbf{Q}, \mathbf{w}, \mathbf{K}$ defined in Section 5. There is a Borel map $F : \mathbf{W} \rightarrow \mathbf{X}$ such that*

$$\mathbf{K}(w, \cdot) = \delta_{F(w)} \quad \text{for } \mathbf{w}\text{-almost every } w, \quad X = F(W) \quad \mathbf{Q}\text{-almost surely.}$$

For each fixed $g \in \text{Aut}(T_d)$, $F(gW) = gF(W)$ almost surely. The free Ising law is the law of a Borel map of vertex IID uniform labels which commutes with every automorphism on every label input. The same spin-factor conclusion holds at $\theta = 0$.

Proof. First suppose $\theta > 0$. The two uniqueness propositions give $V(t) = 0$ for every deterministic time in their respective parameter ranges. Invariance permits any vertex in place of the auxiliary vertex o . Intersecting the full-probability events for all vertices and nonnegative rational times gives $h_v(q) = h'_v(q)$ simultaneously at those indices. Both field arrays are continuous, so they agree at every time and vertex. Hence their magnetizations agree, and the common innovation identity gives

$$X_v(t) = W_v(t) + \int_0^t m_v(s) \, ds = W_v(t) + \int_0^t m'_v(s) \, ds = X'_v(t).$$

Thus $\widehat{\mathbf{Q}}\{X = X'\} = 1$ as an equality of full path configurations. The use of rational times avoided an uncountable intersection over time.

For $v \in \mathcal{V}$ and $q \in \mathbb{Q}_{\geq 0}$ choose the finite Borel version

$$f_{v,q}(w) = \int x_v(q) \mathbf{K}(w, dx)$$

when the absolute integral is finite, and set it to zero otherwise. Kernel integration makes this Borel, and integrability of $X_v(q)$ ensures that the exceptional inputs have w -measure zero. Taking expectations in (5.4) and using $X = X'$ gives

$$0 = \mathbb{E}_{\mathbb{Q}}(X_v(q) - X'_v(q))^2 = 2\mathbb{E}_{\mathbb{Q}} \text{Var}(X_v(q) \mid W).$$

It follows that $X_v(q) = f_{v,q}(W)$ almost surely. We may impose these identities simultaneously for all the countably many pairs (v, q) . Disintegrating them shows that, for w -almost every w , $K(w, \cdot)$ gives mass one to

$$\{x \in \mathbf{X} : x_v(q) = f_{v,q}(w) \text{ for every } v, q\}.$$

This set has at most one element, since continuous paths are determined by their rational-time values. Its conditional mass one also makes it nonempty. Thus $K(w, \cdot)$ is a point mass for almost every w .

The conditional point-mass principle for measurable recovery also appears in Kurtz [11, Lemma 1.3 in arXiv:1305.6747v2]. We give a Borel choice of that path, including values on all other inputs. For each v and integer $j \geq 1$, let $p_{v,j}(w)$ be the piecewise linear path whose value at $k2^{-j}$ is $f_{v,k2^{-j}}(w)$ for every integer $k \geq 0$. These are Borel maps to $C([0, \infty), \mathbb{R})$ with its locally uniform topology. Let $\mathcal{G}$ be the set of w for which, at every vertex, these paths are Cauchy uniformly on every compact interval and $f_{v,0}(w) = 0$. This set is Borel. Indeed, the uniform Cauchy condition on $[0, N]$ can be tested by the supremum over rational times there, and the quantifiers over vertices, integers N , error levels, and pairs of sequence indices are all countable. On $\mathcal{G}$ the limits are continuous paths starting at zero. Define $F_v(w)$ to be this limit for $w \in \mathcal{G}$, and let $F(w)$ be the all-zero path configuration for $w \notin \mathcal{G}$. The limit map on the Cauchy set of a Polish space is Borel, so this defines a Borel map $F : W \rightarrow \mathbf{X}$.

For almost every w , the rational coordinates just specified are those of the unique path configuration charged by $K(w, \cdot)$. Uniform continuity of each such path on compact intervals shows that its dyadic linear interpolants converge uniformly there. Consequently $w \in \mathcal{G}$ and $F(w)$ is that configuration. This proves both $K(w, \cdot) = \delta_{F(w)}$ almost everywhere and $X = F(W)$ in the original law.

Fix now one automorphism g . Invariance of $\mathbb{Q}$ and uniqueness of conditional expectations imply

$$f_{v,q}(gw) = f_{g^{-1}v,q}(w) \quad \text{for } w\text{-almost every } w \quad (8.18)$$

for each v, q . Intersecting over those countably many pairs gives all these identities simultaneously for this fixed g . The dyadic interpolants then obey $p_{v,j}(gw) = p_{g^{-1}v,j}(w)$ for every v, j . On this full-measure set, $w \in \mathcal{G}$ if and only if $gw \in \mathcal{G}$, because its defining Cauchy conditions are the same after permuting vertices. The all-zero default is invariant as well, so $F(gw) = gF(w)$ almost surely. The full-measure set in this argument is allowed to depend on g ; none is asserted for all automorphisms at once.

Define a spin map on every innovation input by

$$H_v(w) = \begin{cases} \lim_{n \rightarrow \infty} \frac{F_v(w)(n)}{n}, & \text{if this limit exists and belongs to } \{-1, 1\}, \\ 1, & \text{otherwise.} \end{cases} \quad (8.19)$$

Existence of either indicated limit is a Borel condition expressed through countably many error bounds and integer indices. Thus each H_v is Borel, and so is H into the countable spin product. On the planted probability space,

$$\frac{F_v(W)(n)}{n} = \frac{X_v(n)}{n} = \sigma_v + \frac{B_v(n)}{n}$$

simultaneously for the integer indices. For every $\varepsilon > 0$, the Gaussian tail bound gives

$$\sum_{n \geq 1} \mathbb{P}(|B_v(n)| > \varepsilon n) \leq \sum_{n \geq 1} 2e^{-\varepsilon^2 n/2} < \infty.$$

Borel–Cantelli, first for positive rational ε and then over the countable vertex set, gives $B_v(n)/n \rightarrow 0$ simultaneously for every vertex. Therefore $H(W) = \sigma$ almost surely. Under product Wiener input, H has exactly the free zero-field Ising law. For the fixed g above, the identity $F(gw) = gF(w)$ makes the sequences in (8.19) agree after the same permutation. Their limits and their failure cases consequently agree. The same default sign at every site gives $H(gw) = gH(w)$ almost surely.

Finally, Wiener measure is a Borel probability measure on the Polish space $C_0([0, \infty), \mathbb{R})$. The standard Borel sampling theorem therefore supplies a Borel map $\xi : [0, 1] \rightarrow C_0([0, \infty), \mathbb{R})$ which sends Lebesgue measure to Wiener measure. For independent uniform labels (U_v) , set

$$\Phi(U) = H((\xi(U_v))_{v \in \mathcal{V}}).$$

Applying the same ξ at every site gives product Wiener law and commutes pointwise with every vertex permutation. Thus Φ is Borel, has law $\mu_{d,\beta}$, and satisfies $\Phi(gU) = g\Phi(U)$ almost surely for each fixed g . Lemma 8.5 replaces Φ by a Borel map with the same law that is equivariant on every label input.

When $\theta = 0$, the broadcast law is already a product of fair signs. The rule $\Phi_v(U) = 1$ for $U_v \leq 1/2$ and $\Phi_v(U) = -1$ otherwise gives that law and commutes with every permutation for every input. All choices above are for the fixed pair (d, β) . Together with the known converse in Section 1, this proves Theorem 1.1. $\square$

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